Evaluating Dataset Diversity in Media Bias Detection Models

A thesis submitted in partial fulfillment of the requirements for the degree of the Master of Social and Economic Data Science at the University of Konstanz

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Abstract

Datasets play a crucial role in the performance and generalization of machine learning models. While research has increasingly emphasized data-centric approaches and acknowledged their value-laden properties, these have not been properly researched. In particular, the concept of diversity in dataset curation remains loosely defined, often conflated with scale or bias. Offering a clear operationalization of diversity in datasets, this thesis explores the impact of source and topic diversity of fine-tuning datasets on media bias detection models, a challenging natural language processing task due to the subtle and context-dependent nature of bias.

Using Anno-Lexical (a large-scale dataset with synthetic annotations) and BABE (a smaller expert-annotated dataset), we assess how diversity affects model accuracy, robustness, and bias mitigation. Results show that, while diversity does not significantly improve performance on standard metrics, it has the potential to enhance generalization capabilities, particularly in fairness-related evaluations. However, this effect is dataset-dependent: source diversity in Anno-Lexical improves bias mitigation, while topic diversity and the BABE dataset show no consistent benefits.

These findings highlight the importance of context in dataset diversity and contribute to improving dataset curation strategies for NLP tasks, advocating for more precise diversity measurement and evaluation techniques.

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1. Introduction

1.1 Motivation

Datasets play a critical role in the performance and generalization of machine learning (ML) models. While much of the focus in ML research has traditionally been placed on model development, recent advancements emphasize the importance of data-centric practices to improve transparency, reproducibility, and downstream outcomes (Paullada et al., 2020; Sambasivan et al., 2021). Despite these efforts, key concepts underlying dataset curation—such as diversity—are often considered self-explanatory, leading to poor definitions, inconsistent measurement, and conflation with other attributes like scale or bias (Scheuerman et al., 2021; Zhao et al., 2024). This lack of clarity makes it difficult to systematically assess how these concepts could affect ML tasks.

Recent studies on pre-training data for Large Language Models (LLMs) have shifted the focus from sheer dataset size to dataset quality, particularly dataset diversity, as a relevant characteristic for downstream performance and fairness (Chen et al., 2024; Feng et al., 2023; Miranda et al., 2024; Zhao et al., 2023). However, while diversity in pre-training data has received growing attention, its role in fine-tuning datasets for task-specific models remains largely underexplored, leaving a critical gap in understanding its impact in more specialized contexts.

This thesis takes media bias detection as a case study to examine the role of dataset diversity, an inherently challenging natural language processing (NLP) task due to the complexity and subtlety of bias (Lim et al., 2020). Transformer-based models dominate this field, but their reliance on annotated data presents challenges: constructing high-quality datasets is costly, and available datasets are typically small (Rodrigo-Ginés et al., 2024; Spinde, Plank, et al., 2021; Wessel et al., 2023). Recently, advances in

LLMs have enabled the creation of large-scale datasets with synthetic annotations, achieving performance comparable to traditional human-annotated datasets (Horych et al., 2024). These developments present an opportunity to comprehensively investigate the relationship between different dimensions of dataset diversity and model performance. This thesis takes media bias detection as a case study to explore how dataset diversity—particularly in terms of source diversity and topic diversity—affects both model performance and generalization capabilities. In particular, we aim to answer the following research questions:

RQ1. How can diversity in news datasets be measured effectively?

RQ2. Does a more diverse training dataset improve model performance for media bias detection?

RQ3. Do more diverse training datasets present improved generalization capabilities in media bias detection?

We hypothesize that increasing dataset diversity, particularly in terms of source diversity and topic diversity, leads to improved model performance and generalization capabilities in media bias detection.

This research addresses several gaps in the literature by systematically investigating these questions. First, it offers a clear operationalization of dataset diversity, providing a replicable measurement framework applicable beyond this task. Second, while the role of diversity in pre-training datasets has been extensively studied, its role in specific fine-tuning tasks remains underexplored. Our focus on media bias detection tackles this gap, and we further look into two differently curated datasets to obtain deeper insights into this challenge. Finally, by incorporating adversarial testing, we move beyond evaluating overall performance to examine how diversity impacts robustness and generalization.

Overall, we seek to contribute to the data-centric literature by proposing a structured approach to measuring dataset diversity and its downstream effects, which can serve as a guide for future dataset collection strategies.

The source code and supplementary materials for this thesis are available at: https://github.com/SofiaKarsaclian/dataset-diversity.

1.2 Thesis Outline

This thesis is structured as follows. Chapter 2 provides the theoretical background for diversity measurement, reviewing its key properties and methods widespread across disciplines. It also looks at related work in the ML field and explores media bias detection as a case study, highlighting both opportunities and challenges. Chapter 3 outlines the research methodology used to address the research questions. This includes how diversity is measured, which dimensions to assess and how to quantify them, the datasets utilized, the subsampling strategy adopted, and the experimental setup. Chapter 4 presents the findings, focusing on the diversity of sources and topics within the dataset. Chapter 5 interprets the results, discussing their implications and limitations. Finally, Chapter 6 concludes the thesis with a summary of the key contributions and directions for further work.

2. Related Work

2.1 How is Diversity Measured?

Diversity is an essentially contested concept (Gallie, 1956), meaning that it is value-laden, lacks a fixed definition, is used in different contexts, and involves internal disagreements. In the field of ML, this conceptual vagueness translates into a lack of clear, standardized definitions and validation methods for diversity (Zhao et al., 2024). To understand this challenge better, it helps to examine how diversity has been measured in other fields.

The measurement of diversity has its roots in ecology, where researchers sought to quantify biodiversity—the variety and distribution of species within ecosystems. Over time, this concept migrated into social sciences, economics, and more recently, into ML (Mitchell et al., 2023). Diversity can be loosely defined as an attribute of any system whose elements may be apportioned into categories (Stirling, 2007).

In practice, diversity can be parametrically defined in relation to some empirical or theoretical distribution relevant to the system under study (e.g., species richness in an ecosystem, or ethnic diversity in a population). However, in many cases, non-parametric approaches are necessary, especially when clear parameters are lacking or when a more generalized, transferable measure is needed across different contexts. It is within these non-parametric approaches that Stirling (2007) recognizes a common understanding of diversity that spans disciplines. This conceptualization is based on the combination of three basic properties: (i) variety—the number of distinct categories into which the system elements are distributed, (ii) balance—the relative distribution across categories, and (iii) disparity—the degree of difference between categories.

Each of these elements is a necessary but insufficient property for diversity. Moreover, these concepts are intertwined since each element both defines and is defined by the other two (Stirling, 2007). Even if diversity is measured using only variety and balance (such as with the Gini Index), disparity is still indirectly considered—because the way the elements were categorized and classified already reflects certain assumptions about how they differ.

For example, a system including ten different dog breeds is less diverse than a system including ten entirely different types of animals—such as dogs, butterflies, birds, snakes, and so on. Both systems may have the same variety (ten categories) and balance (an equal number of individuals in each category), but the second system is clearly more diverse because the differences between categories (the disparity) are much larger—for instance, the difference between a dog and a bird is far larger than the difference between a Golden Retriever and a Beagle. The characterization of diversity therefore always depends on perspective and context.

Across disciplines, there is an abundance of indexes to measure diversity, although most of them ignore the property of disparity (Van Dam, 2019). One of the first attempts at measuring diversity comes from E. H. Simpson, who defines diversity as "the probability that two individuals chosen at random and independently from the population will be found to belong to the same group" (Simpson, 1949, p. 688). This measure emphasizes the balance property and is calculated as $\sum_{i=1}^{S} p_i^2$, where p_i is the proportion of individuals in the category i, and S is the total number of categories in the population. This measurement is closely related to others used in different disciplines, such as the Herfindahl-Hirschman Index (HHI) in Economics or the Blau Index in Psychology.

Shannon entropy represents another widely-used diversity measure, particularly in information systems, defined as "the minimum volume of communication required to code a message" (Guevara et al., 2016, p. 69). Calculated as -$\sum_{i=1}^{S} p_i \ln p_i$, Shannon entropy is sensitive to both variety and balance, giving more weight to rare categories than Simpson's index. Hill numbers extend this approach by creating a parametric family of diversity indices, defined as $\left(\sum_{i=1}^{S} p_i^q\right)^{\frac{1}{1-q}}$, where the parameter q determines the sensitivity to abundance. When $q = 0$, it measures only variety. When $q = 1$, it corresponds to the exponential of the Shannon entropy, which emphasizes both variety and balance (Hill, 1973). Moreover, it has been shown that many traditional diversity measures, which were originally introduced based on heuristics, are in fact transformations of Hill numbers (Van Dam, 2019).

Despite these advances in incorporating variety and balance, these measures still do not explicitly account for disparity between categories, treating all categories as equally distinct.

Finally, the Rao-Stirling (Stirling, 2007) measures diversity as the sum of pairwise disparities, weighted in proportion to contributions of individual system elements: $D = \sum_{i,j(i \neq j)} d_{ij} \cdot p_i \cdot p_j$, where p_i and p_j represent the relative frequencies of elements i and j in the system (capturing balance), while d_{ij} quantifies the degree of dissimilarity between these elements. In this way, this formulation considers disparity in an explicit way. However, several researchers (Leydesdorff et al., 2019; Van Dam, 2019) have identified conceptual and methodological challenges with this unified approach, arguing that the three properties should be decomposed for a better analysis.

In this study, we address the shortcomings of traditional measurements by employing the Vendi Score (Friedman & Dieng, 2023), a recently developed metric designed to provide a more fine-grained and reference-free assessment of dataset diversity in ML applications, which will be later introduced in Chapter 3.

2.2 Diversity as a Value in Machine Learning Datasets

The capacity to build large datasets from more readily available information online, combined with the reliance on non-expert crowd-sourced workers, has contributed to the development of ML as a suitable path for generating a more neutral and abstracted way of producing datasets (Paullada et al., 2020).

However, it is a mistake to assume that increasing the size of data alone solves the issue of having representative or diverse data (Boyd & Crawford, 2012). Data size does not make it less important to understand the origin of the data in order to account for possible biases and limitations. Even though there has been a large effort in increasing dataset documentation practices in order to increase transparency and make these biases and limitations explicit (Bender & Friedman, 2018; Gebru et al., 2021; Sambasivan et al., 2021), such documentation practices are not always properly able to reflect the values involved in the curation process (Scheuerman et al., 2021). Furthermore, Denton et al. (2020) argue that the role of "working infrastructure" for datasets in ML tends to invisibilize them, leading to their easy naturalization and uncritical adoption within everyday work routines.

Focusing efforts on dataset curation and the examination of the values behind the collection criteria is key for mitigating ethical shortcomings, since datasets determine how the models work and how problems should be framed (Denton et al., 2020).

One of the key values in dataset collection, often assumed to be inherent, is diversity——yet it is frequently overlooked. Zhao et al. (2024) emphasize the importance of conceptualizing, operationalizing, and evaluating diversity in ML datasets to assess whether the proclaimed diversity by dataset curators is in fact achieved. This is particularly crucial given the contested nature of the concept, as differences in operationalization often stem from underlying theoretical disagreements about what diversity entails (Jacobs & Wallach, 2021). Diversity is closely linked to representativeness, bias (Cai et al., 2022), and coverage (Mironov & Prokhorenkova, 2024). A lack of representative or unbiased data can hinder a model's ability to generalize effectively (Sambasivan et al., 2021) and increase the risk of learning biased shortcuts (Geirhos et al., 2020).

Diverse datasets can help mitigate certain forms of bias by ensuring the model is exposed to a wide range of examples across the target domain. However, a dataset can be diverse and still contain harmful biases if certain dimensions of diversity are prioritized over others or if the diversity introduced reinforces problematic patterns.

For example, a news dataset might be diverse in terms of the number of news sources (variety), have an even distribution of sources (balance), and include sources with significantly different political perspectives (disparity). However, the dataset could still be biased among other dimensions, such as the topic of the articles, the time of publication, or the subject of interest. This highlights the importance of considering which dimensions of diversity are relevant for a given task and context.

2.3 Diversity in Pre-Training Datasets

In recent years, with the rise of LLMs, increasing attention has been paid to the composition and diversity of pre-training corpora for these models. While the impact of model and dataset size on performance has been well studied, aspects of data quality——such as bias, toxicity, and diversity——have received comparatively less attention, largely due to the challenge of measuring these complex and interrelated factors. However, recent studies suggest that data quality may be more important than data quantity for downstream performance (Chen et al., 2024).

This focus on quality aspects becomes even more relevant given that the total effective stock of public human text data in the web is expected to be exhausted within this decade (Villalobos et al., 2024). While advancements in LLMs have heavily relied on expanding pre-training size, future improvements will need to shift their focus on optimizing dataset quality and diversity.

To estimate the impact of diversity in pre-training as well as enhance diversity within existing datasets, several approaches have been proposed (Gong et al., 2019). Miranda et al. (2024) use the Task2Vec Diversity Coefficient to measure the variability of the text data pre-training datasets, finding that performance improves with diversity. On the other hand, Zhao et al. (2023) estimate class diversity (words) with the size of the least singular value of the last linear layer in pre-training, and show it improves the statistical efficiency of downstream tasks. Similary, Tirumala et al. (2023) further highlight the importance of careful data selection by introducing D4 (*Document*

De-Duplication and Diversification), a method for data curation that seeks to diversify its distribution by clustering embeddings through k-means. Their results show that D4 accelerates training while boosting average accuracy across 16 NLP tasks. Longpre et al. (2023) also investigate how dataset design choices affect model performance, highlighting that drawing from diverse sources—such as books, news, and web content—yields the greatest benefits.

All these approaches to measuring diversity are usually parametric, meaning they either construct theoretical upper and lower bounds, or rely on maximally diverse datasets for comparison.

Diversity and quality in instruction-tuning datasets has also been shown to be correlated with improved downstream instruction-following performance (Wang et al., 2024; Zhou et al., 2023). Zhou et al. (2023) introduce the idea that *less is more for alignment*: they fine-tune a 65B parameter LLaMA model using only 1,000 carefully curated, diverse prompts, optimized through cluster sampling. Despite their small instruction set, their model achieves competitive performance against state-of-the-art models like GPT-4 and Claude, suggesting that limited (high-quality) tuning data is needed to achieve good outputs.

While diversity in pre-training has been extensively studied, diversity in fine-tuning datasets contexts remains underexplored. High-quality fine-tuning datasets for complex tasks such as media bias detections are generally costly and labor-intensive. This consideration motivates our investigation into how diversity can be leveraged to curate more efficient and effective training data for such tasks.

2.4 Media Bias Detection

Media bias can broadly be defined as the deliberate slanting of information by journalists to influence the receiver's opinion in a particular direction (Rodrigo-Ginés et al., 2024). Media bias differs from the expression of mere opinion in that it involves the unfair distortion of information to promote a specific viewpoint. According to the literature, media bias is typically characterized by three key aspects: it is slanted toward a particular perspective, it represents a systematic tendency rather than an isolated incident, and it often involves creating "memorable" or "spin" stories. While most scholars argue that media bias is intentional and influential, others suggest it can also emerge unintentionally.

Although some degree of bias is inevitable, it is crucial for readers to be able to recognize and critically assess media bias in order to mitigate its consequences. Unchecked bias can contribute to the formation of filter bubbles and the development of narrow and one-sided understandings of complex issues, which can in turn influence voting behavior (Spinde et al., 2024).

Media bias can manifest at various stages of the news production process and through different types of distortion. To provide a systematic overview, Spinde et al. (2024) developed the *Media Bias Taxonomy*, synthesizing insights from existing literature to organize the different definitions of media bias.

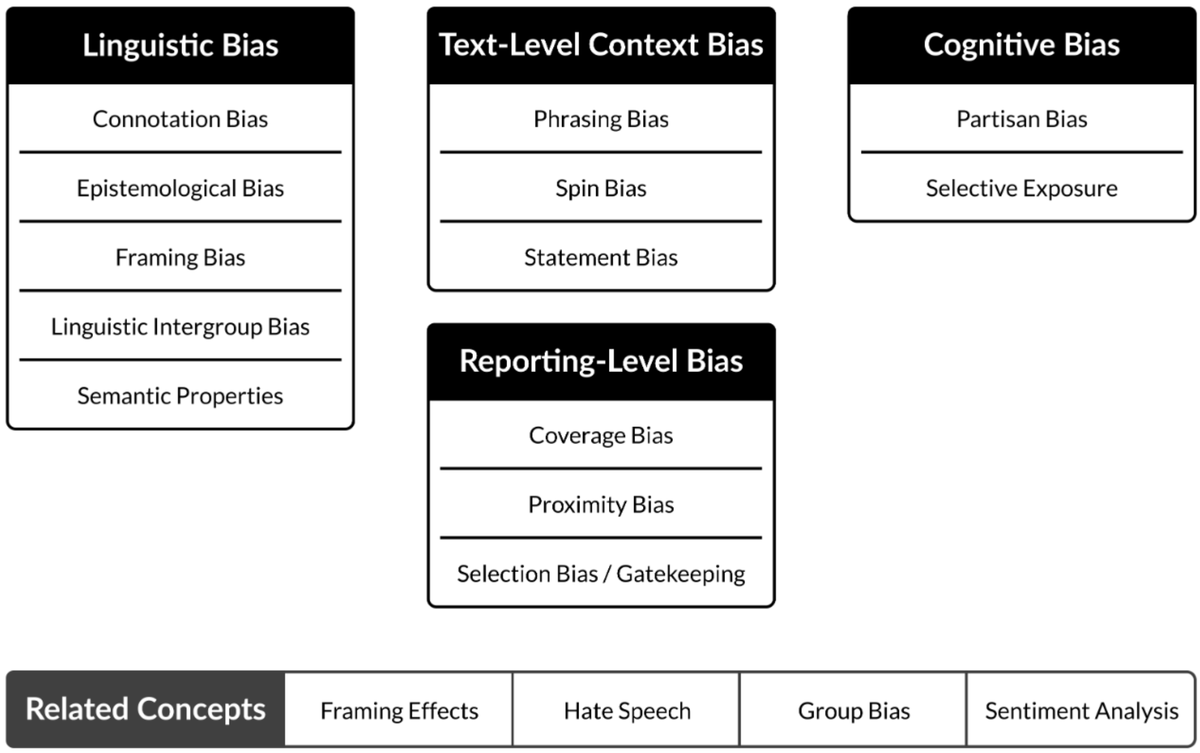

Figure 2.1: The Media Bias Taxonomy (Spinde et al., 2024)

In this study we focus specifically on linguistic bias, also referred to as lexical bias, which Spinde et al. (2024) define as "a pattern of using certain words that reflects a particular way of thinking about a group or an individual based on their social category" (p. 7). Linguistic bias encompasses all forms of bias induced by lexical features, such as word choice and sentence structure (Wessel et al., 2023).

Detecting bias in media is an inherently challenging NLP task due to the subtle and context-dependent nature of bias. Transformer-based models, such as BERT and RoBERTa, dominate this field, relying heavily on annotated datasets (Spinde, Plank, et al., 2021). Its reliance on fine-tuning datasets offers them an advantage over zero-shot LLMs given the subjective nature of bias perception (Wen & Younes, 2023). However, annotated datasets for media bias detection are often limited in size and scope, which can hinder model generalization.

Recent research highlights the potential of large-scale pre-training and synthetic annotation techniques to address these limitations (Horych et al., 2024). Despite these advancements in dataset scale, there is still limited understanding of the role of dataset diversity in model performance. Although Horych et al. (2024) emphasize the importance of a well-balanced corpus in terms of data sources across the political spectrum to improve the models and reduce potential biases, a more thorough examination of how diversity impacts these models is still needed.

2.5 Summary

This chapter explored the theoretical foundations of diversity measurement, its application in machine learning contexts, and its specific relevance to media bias detection. Several insights emerge from this review. Diversity reveals itself as a multidimensional concept best understood through the properties of variety, balance, and disparity, though most measurement approaches fail to account for all three dimensions simultaneously. Although its effects are hard to isolate, the role dataset diversity plays within machine learning is significant, being able to affect model generalization, fairness, and overall performance for pre-training datasets. However, its role in fine-tuning datasets for specific tasks like media bias detection remains underexplored.

While it has been critical to the success of language models, dataset size does not guarantee quality. This issue becomes relevant for tasks such as media bias detection datasets in the era of scalable synthetic annotation methods. Media bias detection stands out as a particularly complex NLP where dataset diversity may be especially influential due to the inherently subjective and multifaceted nature of bias itself. In many cases, uncurated or imbalanced data can reinforce biases and limit a model's ability to generalize. This makes dataset diversity—alongside other quality considerations—an essential factor deserving further study.

These insights collectively point towards the need for more nuanced approaches to diversity measurement and optimization in specialized NLP applications. By shifting the focus from data quantity to dataset quality, the field can work toward more robust, fair, and generalizable machine learning models.

3. Methodology

Building upon the foundation laid in Chapter 2, we investigate the impact of diversity on automatic media bias detection. First, we define how to quantitatively measure diversity in the context of this task and discuss several dimensions across which this could be measured. Then we proceed to do so across two dimensions that we deem important for this specific task: source and diversity. We measure diversity in two datasets created for the same task: Anno-Lexical and BABE. We experiment with creating same-size subsets with different diversity levels for each dimension and dataset by using the Dirichlet distribution as a sampling strategy. In line with Spinde, Plank, et al. (2021), we fine-tune a RoBERTa model on each of these subsets. We assess model performance using standard evaluation metrics and extend the analysis with an adaptation of the CheckList-based adversarial tests for media bias (Ribeiro et al., 2020; Wessel & Horych, 2024).

3.1 The Vendi Score

The Vendi Score (Friedman & Dieng, 2023) is a recently proposed metric designed to quantify the diversity of datasets in ML. It was introduced to overcome the limitations of traditional diversity measures, which often rely on a reference distribution. The Vendi Score, in contrast, captures the inherent diversity within the dataset itself, without the need for external references, making it adaptable to different contexts. It is a non-parametric measure that uses the exponential of the Shannon entropy of the eigenvalues from a similarity matrix. The score is mathematically defined as:

$$VS_k(x_1, \ldots, x_n) = \exp\left(-\sum_{i=1}^{n} \lambda_i \log \lambda_i\right) \tag{3.1}$$

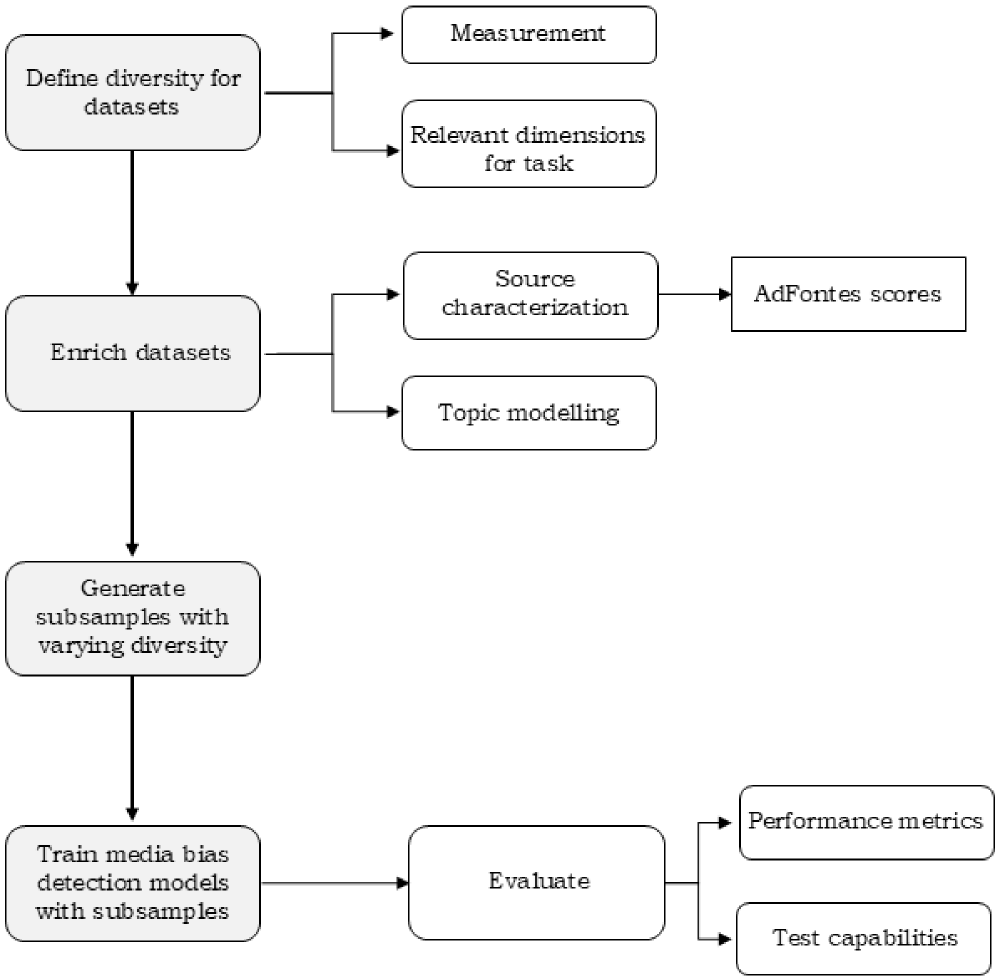

Figure 3.1: Workflow for examining the impact of diversity in automatic media bias detection

Where λ_i represents the eigenvalues of the similarity matrix, and $(x_1, \ldots, x_n)$ are the data points or elements in the dataset. The summation calculates the Shannon entropy over the eigenvalues, and the exponential function transforms this entropy into the final Vendi Score. This score can be intuitively interpreted as the effective number of dissimilar (or unique) elements in a sample.

The similarity matrix is constructed using a user-defined similarity function, which makes the measure adaptable to a wide range of fields.

One of the core strengths of the Vendi Score is its ability to account for the three key properties of diversity identified by Stirling (2007): variety, balance, and disparity. These properties are implicitly incorporated into the measure through its calculation method.

Variety is reflected in the number of distinct features or elements within the dataset, as the diversity measure captures the total number of unique aspects present. Balance is considered given that the Vendi Score treats each item with a level of sensitivity proportional to its prevalence.[1] Finally, disparity is represented by the differences among the features, which is directly captured by the eigenvalues of the similarity matrix used in the calculation.

3.2 Dimensions of Diversity

What aspects are relevant to consider for measuring diversity in the context of media bias detection? To explore this, we first look at the taxonomy induced by Zhao et al. (2024), which identifies several aspects of dataset curation practices in ML. This taxonomy can be adapted to reflect the unique challenges posed by media bias detection tasks, particularly when working with text data from news sources.

Table 3.1 outlines how the diversity dimensions identified by Zhao et al. (2024) can be mapped to characteristics of news corpora and adapted to the needs of this study.

We can observe that diversity is then referred to as variety in some cases (for composition, source and domain), but as representativeness in others (referred to human attributes). However, representativeness must naturally be defined with respect to a reference distribution and must therefore be contextualized for each application.

To start our exploration of diversity and its impact on model performance, we chose to focus on two key dimensions: source and topic. These dimensions were selected based on their potential to influence the text style, terminology, and tone of the data, all of which are critical factors affecting model performance. Different news sources often exhibit distinct writing styles, tone, and formalities, which may pose challenges to the model's generalization capabilities. Additionally, the topic of a news article strongly determines the vocabulary used, ranging from highly technical terms in specialized domains to more

[1] Pasarkar and Dieng (2024) further introduce a family of Vendi Score metrics with different levels of sensitivity to rare or common items, suitable for settings with significant imbalance in the items distribution.

Dimension	Definition	Adaptation for news corpora
Composition	Variety in what a dataset instance contains.	Lexical complexity Vocabulary Language
Source	Variety in where data instances are collected from, such as web source origin or geographic origin.	Social media (publishing account) News outlet (reliability, political leaning) Geographical coverage: regional vs. national, international
Domain	Variety in the topic area of the data instances.	Topic Publication date
Subject	Representation of human subjects in the dataset, such as by protected attributes, physical characteristics, nationality, socioeconomic status, and language.	Named entities
Annotator	Representation of annotator backgrounds.	Demographic background Domain expertise Political affiliation

Table 3.1: Adaptation of diversity dimensions taxonomy for news corpus

general language in everyday topics. By analyzing these two dimensions, we aim to assess whether the model's performance is consistent across varying sources and topics. This approach aligns with the project's scope, and allows for a focused yet meaningful analysis of how diversity can influence model performance.

Depending on the goal of the diversity analysis, the task at hand and its context, and the available metadata, other dimensions (and different degrees of disaggregation for each of them) may be more relevant to examine. Although the Vendi Score provides a useful metric for comparing diversity across various datasets, this comparison is only valid when assessing diversity within the same dimension, which must be defined and constructed consistently.

3.3 Data

The Vendi Score provides an intrinsic measure of diversity for each dataset and can potentially be compared to other training datasets if the tasks and contexts at hand are similar. In this case, we focus on the binary classification of lexical bias at the sentence level. This focus is motivated by the release of a new large-scale (48,330 synthetically annotated sentences) dataset for this task: Anno-Lexical (Horych et al., 2024). The large size of this dataset was enabled by using LLMs as annotators. In a near-unsupervised setting, the dataset was annotated by three top-performing models: Zephyr 7B Beta, OpenChat 3.5, and LLama 2 13B Chat. Models fine-tuned on this dataset achieve performance levels comparable to those annotated by experts. Furthermore, the dataset curation focused on balancing the data sources across the political spectrum, aiming to reduce potential biases.

Previous datasets focusing on the binary classification of lexical bias include BABE—Bias Annotated by Experts—–(Spinde et al., 2021) and BASIL (Fan et al., 2019). As Anno-Lexical, these were also collected from U.S. news outlets, but are significantly smaller in size (4,121 and 7,919 annotated sentences, respectively) and cover fewer sources and topics. BABE covers sentences from 12 predefined controversial topics published on 14 US news platforms from January 2017 until June 2020 (Spinde, Plank, et al., 2021), while BASIL covers all sentences from 100 sets of articles, each set discussing the same event, from three different outlets: *Fox News*, *The New York Times*, and *Huffington Post*. They selected 10 sets for each year from 2010 to 2019. Although our initial intention was to analyze diversity across all three datasets, BASIL proved to be less suitable for a thorough exploration of diversity. The dataset's lexical bias is notably sparse, with only 5% of the sentences containing lexical bias, and there is a lack of annotated examples for this specific type of bias as noticed by Maab et al. (2023).

Therefore, this master thesis focuses on studying the impact of diversity using Anno-Lexical and BABE, combining the advantages of both synthetic and human annotations. This approach allows for a comprehensive examination of how diversity influences model performance, incorporating both the scalability of LLM-generated annotations and the high-quality, expert-annotated data from BABE. While the primary focus will be on analyzing intrinsic dataset diversity, this setup also facilitates a comparison between the two datasets, offering insights into how their inherent diversity affects model outcomes.

3.4 Enriching the Datasets and Diversity Quantification

To perform a comparable analysis, it is essential to maintain consistency in how the dimensions are characterized. Since both datasets focus on U.S. media, we utilize a publicly available *media bias chart* that characterizes the news landscape in the country. Additionally, while the BABE dataset provides information about the topic of the article from which each sentence is extracted, we choose to identify the topic at the sentence level. This approach enhances accuracy and allows for better comparability across different datasets. In the following section, we provide a detailed description of how both dimensions are constructed and measured.

3.4.1 Source characterization

The rising political polarization in the U.S. over the past decade, reflected in the media, has led to the emergence of several fact-checking and media-bias characterization initiatives aimed at identifying the biases of various news outlets. For example, AllSides[2] maintains a bias chart where news outlets are categorized into five groups: left, left-center, center, right-center, and right. They use this framework to present multiple versions of the same news story. Similarly, Ground News[3] evaluates news outlets based on their political bias, reliability, and ownership, merging articles covering the same story to provide a broader range of perspectives. Ad Fontes,[4] a public-benefit corporation established in 2018, regularly evaluates news outlets on a two-point methodology: political bias and reliability. To generate these scores, expert annotators

[2]https://www.allsides.com/unbiased-balanced-news
[3]https://ground.news/
[4]https://adfontesmedia.com/interactive-media-bias-chart/

rate prominent featured articles from each outlet over several news cycles. A politically balanced panel of at least three analysts discusses their ratings and averages their scores. For reliability, the key metrics are expression, veracity, and headline and graphic accuracy. For bias, the focus is on political position, language, and comparison (Ad Fontes Media, 2024).

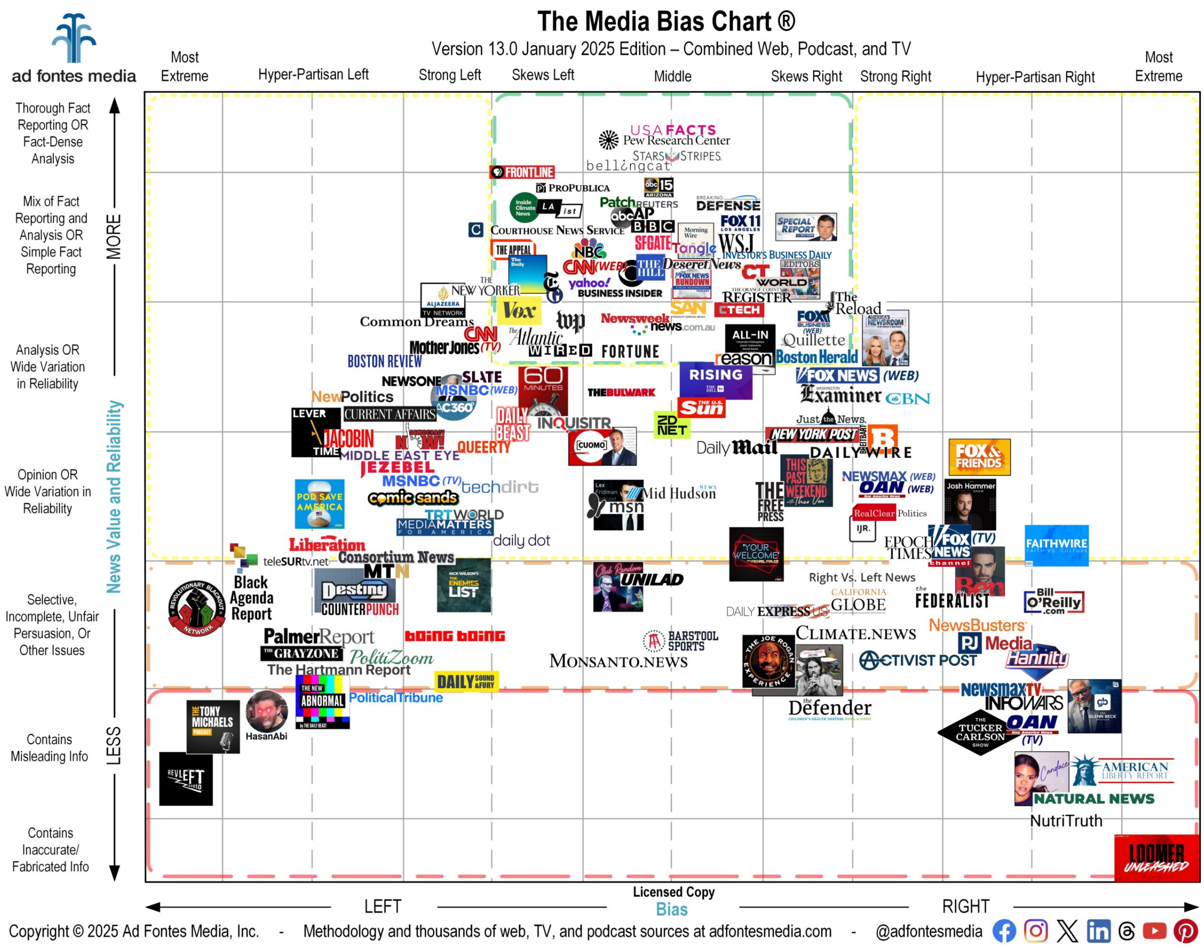

Figure 3.2: The Ad Fontes Media Bias Chart[5]

To generate the similarity matrix for the diversity calculation we rely on the two axes of the AdFontes Media bias chart: reliability and political bias, as a way to characterize the different sources. The advantage of using AdFontes over other media bias charts is that it employs a continuous scale for both dimensions, offering a more precise way to measure disparity across sources. We utilize Manhattan distance to quantify this disparity. Figure 3.3 illustrates how this works by visualizing the similarity between a random sample of 10 sources. A "penalty" is applied to the similarity score for sources that are more similar to each other in their reliability and bias scores (e.g., *The Guardian* and *The New Yorker*, or *The Christian Post* and *Fox Business*). In the case of this sample, which contains 10 sentences from 8 different sources, the effective number of sources is calculated to be 7.39.

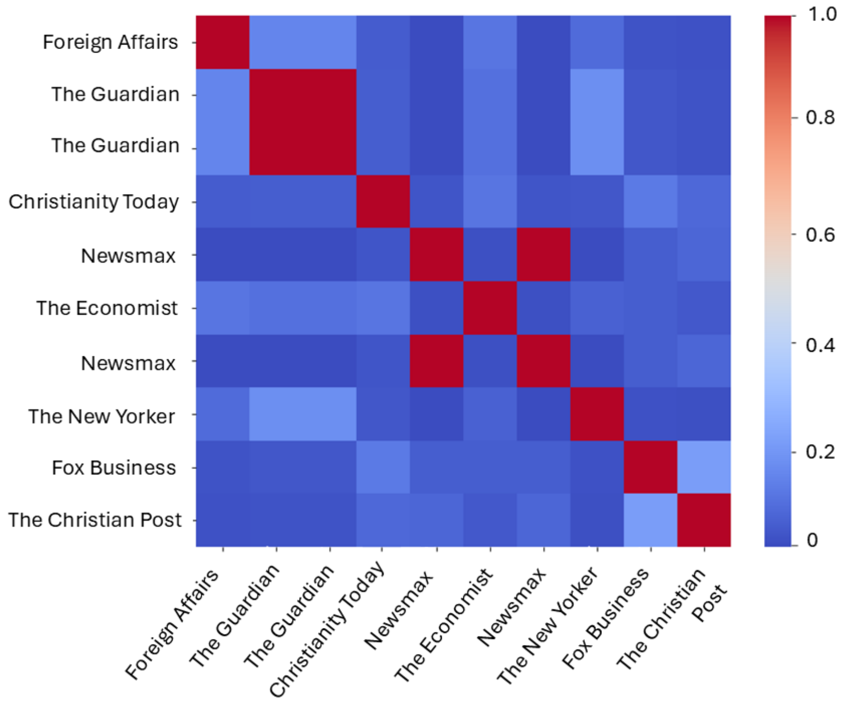

Figure 3.3: Heatmap of similarity between sources

Despite the usefulness of these scores, there are limitations to using them for source characterization. For instance, they may not always reflect the complexity of each outlet's stance. For example, if an outlet publishes many articles with both extreme left and extreme right biases, it may be classified as "center" (implying also that the center is somehow unbiased). Ganguly et al. (2020) have shown that news articles do not always align with the political leaning of the source, and this alignment can shift depending on the topic being reported.

Nevertheless, the Ad Fontes methodology provides a consistent and transparent approach for rating sources, allowing us to identify broad differences between them and quantify disparity. Figure 3.4 illustrates the source distribution for both datasets characterizing them with the AdFontes framework, revealing a stark contrast between the number of sources in Anno-Lexical (118) and BABE (10), as well as between their spread along the political bias and reliability axis.

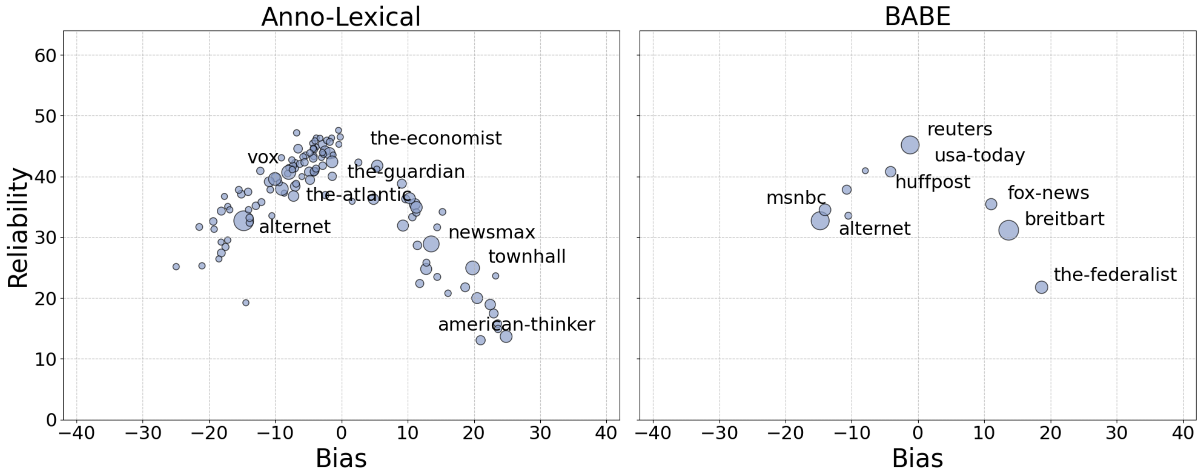

Figure 3.4: Source distribution by bias and reliability for Anno-Lexical and BABE. Circle size represents the relative contribution of each source to its respective dataset, with larger circles indicating more articles from that source. The eight most prominent sources in each dataset are labeled.

3.4.2 Topic Modeling

To characterize the topic diversity of each datasets, we employ state-of-the-art topic modeling techniques. While numerous topic modeling approaches exist in the literature, BERTopic Grootendorst (2022) stands out as a particularly effective method due to its strong performance with improved computational efficiency.

BERTopic addresses the limitations of traditional topic modeling techniques, such as Latent Dirichlet Allocation (LDA), which rely on bag-of-words representations. Instead, BERTopic utilizes pre-trained transformer-based language models, like Sentence-BERT (SBERT), to create semantically rich document embeddings that capture the contextual meaning of texts. These embeddings are then clustered using HDBSCAN, a hierarchical density-based clustering algorithm, to group similar documents together. The resulting topics are represented using a class-based variation of Term Frequency-Inverse Document Frequency (TF-IDF), which considers the relevance of words within a cluster of documents, rather than individual documents. This method enhances the coherence of the generated topics and the overall performance of the model (Egger & Yu, 2022).

One of the major advantages of BERTopic is its flexibility. The separation of document embedding and topic representation allows for each step to be fine-tuned independently. Additionally, UMAP (Uniform Manifold Approximation and Projection) is employed for dimensionality reduction, which further improves the clustering process by preserving both local and global features of the data. This leads to more meaningful and interpretable topic representations. The class-based TF-IDF method refines topic extraction by considering the distribution of words within clusters, resulting in topics with greater coherence.

For this study, we use the `sentence-transformers/all-mpnet-base-v2` model for Sentence-BERT embeddings, as it has demonstrated superior performance across a variety of tasks. The HDBSCAN algorithm is configured with a minimum cluster size of 25 and a minimum sample size of 20, ensuring meaningful clustering of documents. This configuration, combined with further outlier reduction based on topic-document probabilities, ensured that topics were coherent and distinct for both datasets. Additionally, we incorporate LLaMA 3 in a few-shot learning setting to refine topic labels. LLaMA 3 is prompted with representative topic words and example sentences to generate more descriptive and interpretable topic names, enhancing clarity beyond standard term frequency methods.

The coherence of the identified topics is evaluated using the C_V coherence measure (Röder et al., 2015), which assesses the semantic similarity between top words in a topic. Higher values indicate more interpretable and well-defined topics. Our results show reasonable coherence: $\mu = 0.478$, $\sigma = 0.094$ for BABE, and $\mu = 0.506$, $\sigma = 0.193$ for Anno-Lexical, suggesting that the extracted topics maintain meaningful structure. BABE provides an opportunity to evaluate BERTopic's performance, as it includes the annotated article topic from which the sentence was extracted. Figure 3.5 illustrates the topic co-occurrence between the assigned topic at the sentence level by BERTopic and the annotated article topic from which the sentence was extracted, showing a high co-occurrance in most topics.

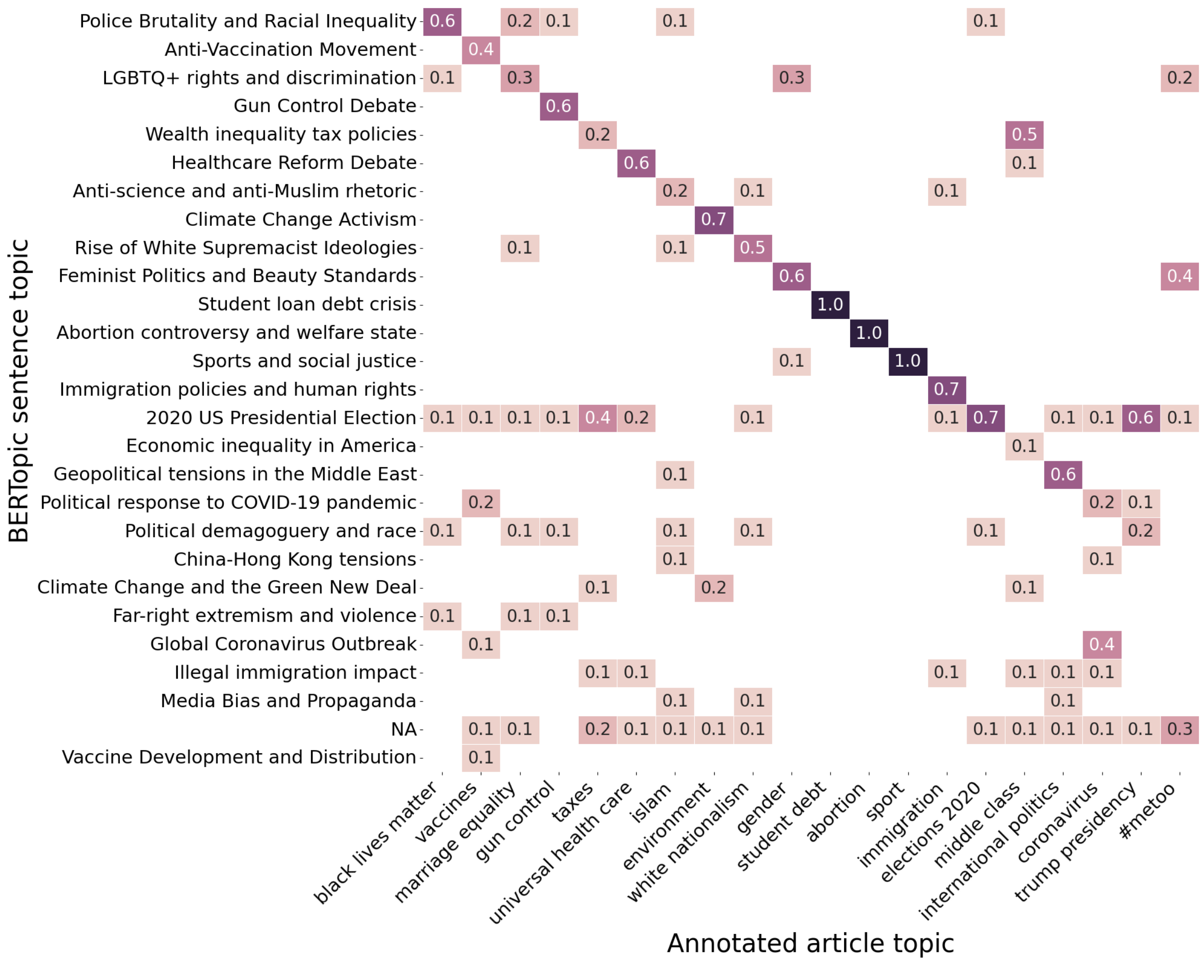

Figure 3.5: Heatmap of topic co-occurrences. Values represent the proportion of sentences with each BERTopic label within articles of a given annotation category, rounded to one decimal place. Darker colors indicate a stronger association.

Some topics such as the "2020 US Presidential Election" show a high co-occurrence with several article-level topics. This is not necessarily problematic, as articles (and in less degree, sentences) can be related to more than one topic. In fact, sentence-level topic identification is often more accurate than article-level classification because of this flexibility, as illustrated in Table 3.2.

To measure disparity between the topics, we use the class-based TF-IDF (c-TF-IDF) matrix, which is particularly well-suited for this task. The c-TF-IDF assigns relevance scores to words based on their frequency within topics relative to the entire corpus, making it an effective tool for assessing topic similarity. Although topic embeddings could also be used for similarity calculations, they are not ideal due to the high dimensionality (358 dimensions) which can lead to less interpretable results. Therefore, the c-TF-IDF matrix

Text	**Annotated Article Topic**	**BERTopic Sentence Topic**
"Video footage on Twitter showed cheering protesters in Raleigh taking down two Confederate statues with ropes and hanging one to a lamp post."	Black Lives Matter	Far-right Extremism and Violence
"While many women, and the men who love them, are well aware that, spurred on by the Vatican, United States Conference of Catholic Bishops (USCCB) and the so-called personhood movement, Republicans are working feverishly to eradicate women's reproductive rights."	Gender	Abortion Controversy and Welfare State
"Defense Secretary Mark Esper has issued a policy ushering in a de facto ban on displaying the Confederate flag at U.S. military installations around the world by authorizing only certain flags that promote unity, according to a copy of his memo, seen by Reuters on Friday."	Black Lives Matter	Far-right Extremism and Violence
"Franklin Graham: Democrats Exploit Coronavirus to 'Defeat Donald Trump in the Election'"	Coronavirus	Political Response to COVID-19 Pandemic
"Harvard University officials said Monday that school will invite only 40 percent of undergraduates, including all first-year students, to campus for the fall semester as the number of coronavirus cases continues to increase in most states."	Marriage Equality	Global Coronavirus Outbreak

Table 3.2: Sample comparison between the annotated topics at the article level and the topic assigned by BERTopic at the sentence level. Even for those that do not match, the predicted topic is coherent.

provides a more practical and interpretable method for measuring similarity across topics.

Following BERTopic's results, BABE contains articles' sentences related to 25 varied controversial political topics. In contrast, the Anno-Lexical dataset contains a much larger variety, containing 120 different topics that extend beyond politics to include a wide range of subjects such as entertainment (music, films), commercial content (advertisements), and lifestyle content (astrology, travel), among many others. Figures A.1 and A.2 in the Appendix display the identified topic clusters for each dataset on a reduced two-dimensional space. There are a number of outlier sentences that do not fall within any of the created topics (7476 in Anno-Lexical and 233 in BABE). This however does not represent an issue for our application, as we can create subsets using only the identified topics.

3.5 Subsampling Strategy

We aim to create multiple subsets with varying levels of diversity, which we can control by adjusting the three main properties previously discussed. To achieve this, we use the Dirichlet distribution, a method that helps account for both balance and variety within the sample. This approach enables us to randomly sample from specific sources or topics while mitigating unintended bias in the subsampling process.

While traditional approaches like simple random sampling or stratified sampling could be considered for this task, they lack the flexibility to systematically control diversity levels across multiple dimensions. Simple random sampling would fail to guarantee representation of minority categories, while standard stratified sampling maintains fixed proportions that don't allow for the controlled variation in balance we require.

The Dirichlet distribution is a multivariate generalization of the Beta distribution and is defined by k positive parameters α_k in a k-dimensional space. In the special case where $k = 2$, the function $f(y_1, y_2)$ represents the probability density function of the Beta distribution with parameters α_1 and α_2 (Lin, 2016). The probability density function of a Dirichlet-distributed random vector X is proportional to:

$$p(x) \propto \prod_{i=1}^{k} x_i^{\alpha_i - 1}$$

where α is a vector containing the k positive concentration parameters.[6] When generating Dirichlet-distributed random samples, we can define a unique α that influences the distribution of the proportions. If $\alpha < 1$, the results will tend to be highly concentrated, often yielding very low values for all but one category. Conversely, if $\alpha > 1$, the resulting values are likely to be more spread out, leading to more balanced and uniform subsamples.

[6]https://numpy.org/doc/2.1/reference/random/generated/numpy.random.Generator.dirichlet

The Dirichlet distribution samples values for each category using Gamma distributions. Each category's value is sampled independently from a Gamma distribution with a shape parameter equal to α and a scale parameter of 1. The resulting samples are then normalized so that they sum to 1.

To create multiple subsets with differing diversity levels, we generate Dirichlet distributions for each dimension (topic and source) by varying the α levels. Although this primarily affects balance, it also has an indirect effect on variety, as highly concentrated distributions tend to significantly reduce the number of categories.

For each dimension, we use 12 different α values, ranging from 0.001 to infinity (where infinity represents the most uniform distribution possible), and perform three iterations for each value. This results in 36 subsamples, each containing 500 samples for each dimension or dataset. We characterize each subset with their corresponding Vendi Score, for source and topic.

The size of the subsamples allows for better control over highly concentrated distributions. For example, to create subsamples dominated by a single topic, we ensure that each topic has more than 500 observations, which ensures meaningful sampling even when the concentration is highly skewed.

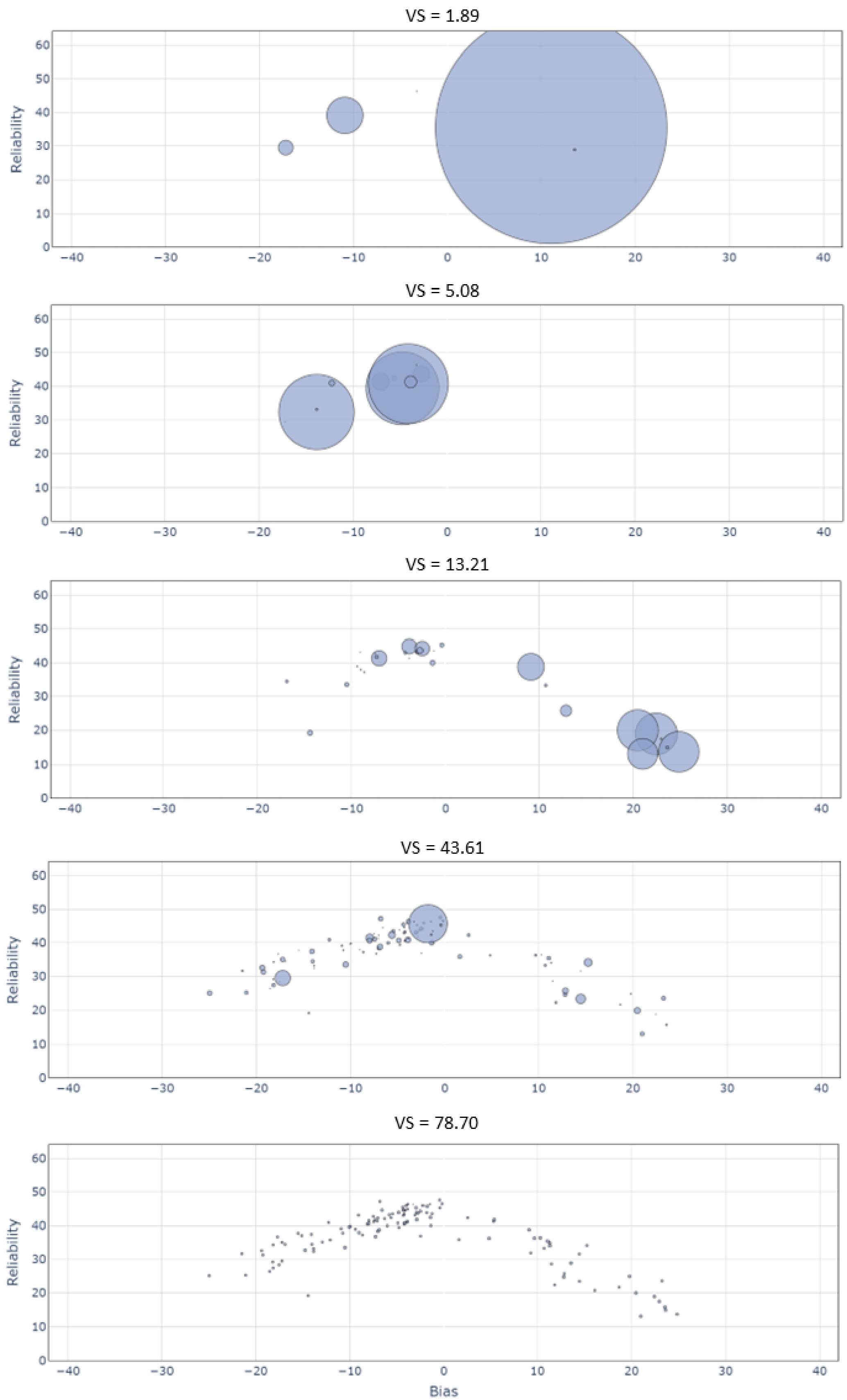

Figure 3.6: Source distribution for five subsets from Anno-Lexical with varying Vendi Score. Bubble size is proportional to the number of the samples from each source (each subset contains in total 500 samples).

3.6 Model Training

Each subset is used to fine-tune a RoBERTa model (Liu et al., 2019). RoBERTa stands for Robustly optimized BERT approach. As its name suggests, it is a transformer model that builds upon the architecture of BERT by optimizing key hyperparameters and training on larger datasets. It was trained on diverse text content, such as the English Wikipedia, Books Corpus, WebText corpus, the CommonCrawl News dataset—containing 63 million articles from 2016 to 2019—, and the Stories from Common Crawl (Liu et al., 2019). Unlike BERT, RoBERTa eliminates the Next Sentence Prediction objective and trains on much longer sequences, which results in improved performance across a range of natural language understanding tasks. It has demonstrated exceptional performance in many NLP applications, including media bias detection (Raza et al., 2024; Spinde, Rudnitckaia, et al., 2021).

We train 144 models in total with each of the subsamples, and test them against a previous split of the BABE dataset (also 500 sentences). To evaluate the performance, we use standard metrics (F1, ROC-AUC, accuracy), as well as the Matthews Correlation Coefficient (MCC). MCC is not as widespread of a metric but it is particularly useful for evaluating binary classification models, as it takes into account all four quadrants of the confusion matrix (true positives, true negatives, false positives, and false negatives) (Chicco & Jurman, 2020). Unlike accuracy, which can be misleading when dealing with imbalanced datasets, MCC provides a more balanced measure of model performance, providing a single score that ranges from -1 (perfect inverse prediction) to +1 (perfect prediction), with 0 indicating no better than random prediction.

The MCC is calculated using the following equation:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{3.2}$$

where TP is the number of true positives, TN the number of true negatives, FP the number of false positives, and FN is the number of false negatives.

3.7 The CheckList Framework

Testing against held-out data is a good indicator of model performance, however, since this data is collected using the same procedures as the training data, it is subject to the same biases. Overreliance on downstream evaluation through benchmarking may lead to an inaccurate assessment of model capabilities by prioritizing performance metrics over a deeper analysis of the dataset's intrinsic characteristics (Zhao et al., 2024).

Models can perform well on sets drawn from the same distribution but fail to do so in out-of-distribution tests, due to using shortcuts as a decision rule (Geirhos et al., 2020). For example, if all sentences mentioning left-wing parties in the training data are biased, a model may learn a simplistic heuristic—associating left-wing content with bias—rather than relying on more meaningful linguistic features such as loaded adjectives and adverbs. Generalization tests for detecting this kind of shortcuts becomes particularly relevant when evaluating training sets with varying levels of diversity, as a lack of diversity can lead models to overfit to spurious correlations rather than learning the underlying patterns necessary for robust bias detection.

CheckList is one of such evaluation frameworks, designed by Ribeiro et al. (2020) to break down model behavior and systematically assess where the models may fail or exhibit undesirable behavior. It introduces three key tests for evaluating NLP capabilities. The Minimum Funtionality Test (MFT) serves as a "sanity check", providing the model with simple and focused examples to check if the model is not using shortcuts for classification instead of mastering the capability. The Invariance Test (INV) involves applying label-preserving perturbations to the inputs, with the expectation that the model's predictions remains stable when semantically irrelevant modifications are made. Conversely, the Directional Expectation Test (DIR) introduces controlled perturbations that should lead to a predictable change in the model's output.

Not all NLP models require the same capabilities, and the CheckList tests are meant to be adapted accordingly. Wessel and Horych (2024) extend this approach to media bias detection, an adaptation further refined by Horych et al. (2024). They build adversarial test sets by creating different perturbations on the BABE dataset. The MFT involves a factual test, where the model is tested with short factual sentences that should always be unbiased. The INV is elaborated to stress-test with perturbations across three different categories: replacing locations with other random locations, replacing named entities with pronouns, and interchanging minority groups. Lastly, the DIR involves the injection of biased adjectives and adverbs that should change the prediction to biased.

Test Type	**Test**	**Examples**
MFT	**Factual test**: Short neutral sentences from fact-checking datasets should be unbiased.	"What is a stereotype? An unfair generalization about a group of people."
INV	**Locations test**: Replacing locations with random locations should not change the label.	"**Hawaii** → **U.S.** eyes even stricter gun laws in wake of shooting that killed 2 police officers."
	Pronouns test: Replacing named entities with He/She/Them should not change the label.	"Despite **Portman's** → **her** insistence that she has tried to advance female directors, only one of her feature films was directed by a female."
	Prejudice test: Replacing one minority with another should not change the label.	"For some people, **Buddha** → **Christ** holds immense significance."
DIR	**Loaded-words test**: Injecting biased adjectives and biased adverbs should change a neutral statement to a biased one.	"The EU has **shockingly** secured up to **outrageous** 400 million doses of AstraZeneca's experimental vaccine."

Table 3.3: Examples from each CheckList test adapted for media bias detection models (Horych et al., 2024).

4. Results

This chapter presents the results for our second and third research questions: whether increased training dataset diversity leads to better-performing media bias detection models, and whether such diversity enhances model generalization capabilities. Building on the methodology presented in Chapter 3, we conducted experiments by training RoBERTa models on subsets of both Anno-Lexical and BABE, while controlling for both source and topic diversity. The results of these experiments are presented in this chapter.

4.1 Model Performance

We first look at the performance of the fine-tuned RoBERTa models by focusing on the MCC metric.

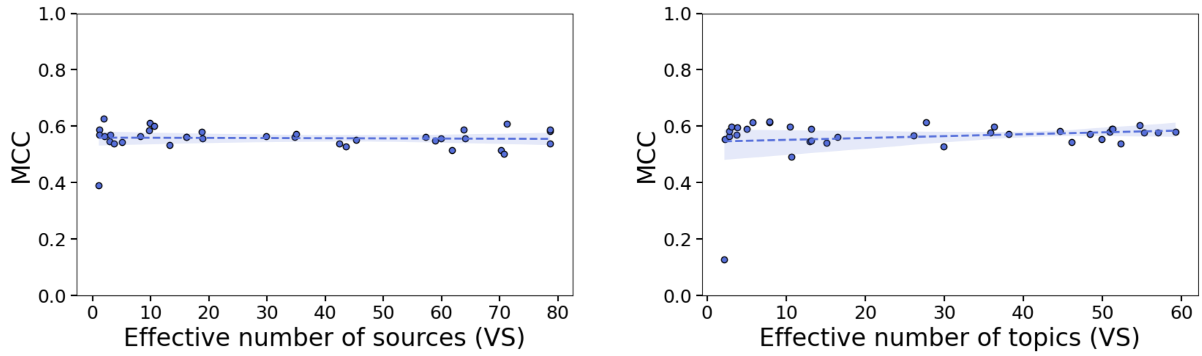

Figure 4.1: Linear regression analysis of the models' MCC against the diversity score (VS) for Anno-Lexical subsets

The models suggest a negligible relationship between the diversity score (VS) and the MCC for both dimensions in the Anno-Lexical subsets. While source diversity has a negligible effect ($\beta_1 = 0, p = 0.823$), for topic diversity the effect is positive but still very small ($\beta_1 = 0.001, p = 0.311$). Neither of the relationships is statistically significant.

The low R-squared values (0.001 and 0.03) indicate that the models are not able to explain the variance in MCC based on the diversity score. This implies that increasing dataset diversity does not necessarily enhance model performance for bias detection, suggesting that other factors, such as annotation quality or linguistic variability, might have a stronger impact.

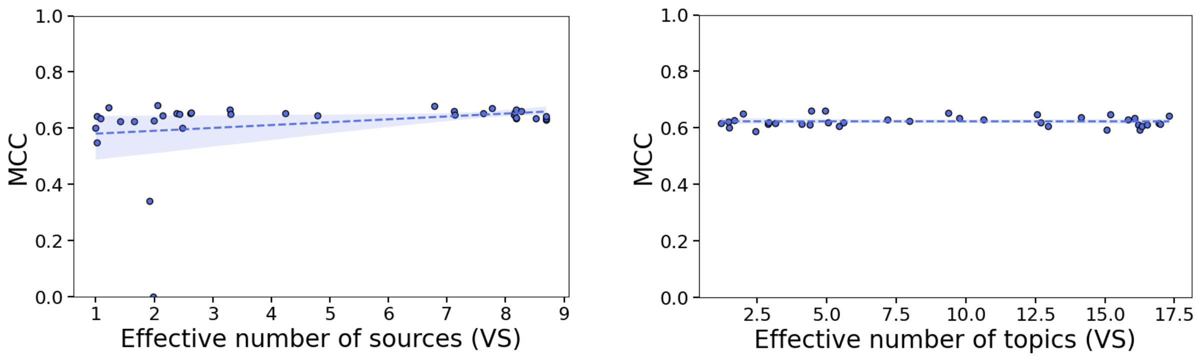

Figure 4.2: Linear regression analysis of the models' MCC against the diversity score (VS) for BABE subsets

This findings also apply for the BABE dataset. Figure 4.2 shows that neither the diversity of sources ($\beta_1 = 0.01, p = 0.129$) nor the diversity of topics ($\beta_1 = 0, p = 0.978$) seem to affect the model's performance.

To further examine the relationship between diversity and model performance, we categorized our training subsets into distinct diversity levels. Rather than treating diversity as a continuous variable, we created binary groupings to compare extremes. Specifically, we applied fixed-width binning to classify subsets into "low diversity" and "high diversity" categories. The low diversity group contains subsets scoring in the lowest 10% of the diversity score range, while the high diversity group includes those with scores exceeding 90% of the maximum diversity score. This approach allows us to directly compare performance between substantially different diversity conditions across each dataset and dimension, potentially revealing threshold effects that might not be apparent in regression analyses.

The boxplot graph indicates that the BABE dataset generally shows slightly higher MCC values compared to Anno-Lexical. This observation aligns with the findings

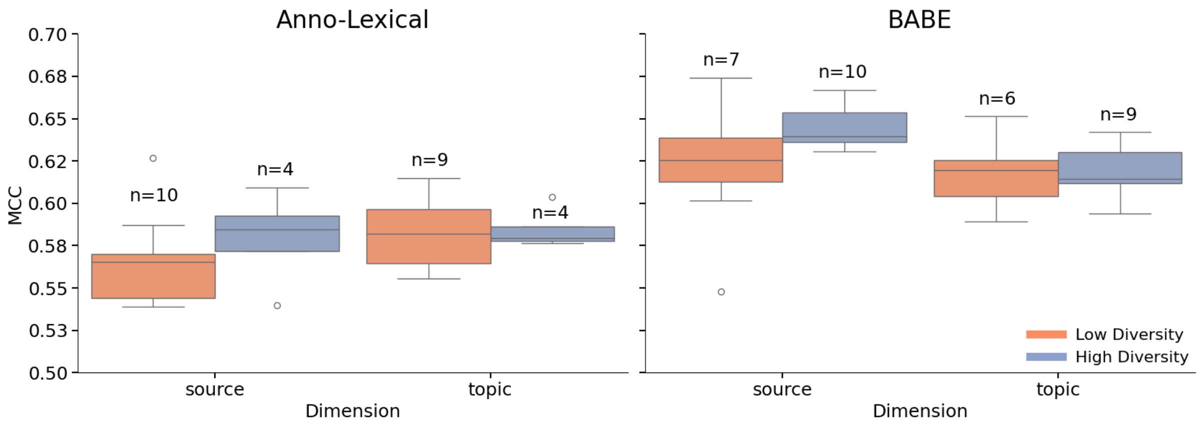

Figure 4.3: Box plots of MCC by diversity dimension and dataset. The y-axis range has been limited to 0.5-0.7 to better emphasize the differences between the main data clusters. Two outliers fall outside this visualization range in the Anno-Lexical subplot (0.39 in the source dimension, 0.13 in the topic dimension).

reported by Horych et al. (2024), who documented similar results when comparing training datasets of equivalent size for both datasets. While visual inspection might suggest potential differences between diversity levels, particularly in the source dimension for both datasets, one-way ANOVAs conducted for each dimension and dataset reveal no statistically significant differences between the low and high diversity groups ($F = 0.791$, $p = 0.391$ for Anno-Lexical source, $F = 0.426$, $p = 0.527$ for Anno-Lexical topic, $F = 3.163$, $p = 0.096$ for BABE source, $F = 0.004$, $p = 0.953$ for BABE topic).

The MCC values across both datasets present a relatively concentrated distribution, as shown by the narrow interquartile ranges and limited number of outliers. This concentration suggests that MCC values remain stable across different diversity levels. Both for Anno-Lexical and for BABE, the high source diversity group appear to have an advantage over the low source diversity group in their performance (with a median of 0.59 vs. 0.57 in Anno-Lexical and 0.64 vs. 0.62 in BABE). However, these differences are not statistically meaningful. This may be partly due to limited sample size rather than conclusive evidence of no effect. Future work with more subsamples could help clarify whether this observed pattern represents a meaningful relationship.

4.2 Generalization Capabilities

We evaluate the model's generalization capabilities through the CheckList framework. We present the results for the three tests adapted by Horych et al. (2024) for media bias: factual, loaded-words, and invariance tests.

4.2.1 Factual and loaded-words tests

Figure 4.4 presents the results of the factual test and the loaded-words test. We now shift our focus to accuracy as the primary performance metric since these tests contain exclusively positive (biased) or negative (unbiased) examples. In these cases, MCC consistently yields null values, making it uninformative for comparative analysis. In contrast, accuracy provides a simple yet effective measure of performance for single-class settings, representing the proportion of correct predictions. The factual test (Minimum Functionality Test) evaluates the model's basic competence by measuring performance on simple factual unbiased examples. The loaded-words test (Directional Expectation Test) assesses whether the model can detect bias when prejudicial or emotionally charged language is introduced. These complementary evaluations provide insight into both fundamental capabilities and sensitivity to linguistic nuance.

In the factual test, subsamples from both datasets demonstrate very high accuracy independently of their diversity levels, with most results exceeding 90%. This suggests that the models can reliably identify simple factual sentences as non-biased. Within the source dimension, diversity levels of the fine-tuning samples do not show a significant effect for either of the datasets. Unexpectedly, within the topic dimensions for BABE subsets, greater topic diversity slightly decreases performance in the factual test ($\beta_1 = -0.003, p < 0.001$). Specifically, the accuracy difference between samples containing only one topic and those encompassing all 25 topics in a balanced manner is estimated at -7.5%. In contrast, topic diversity does not seem to affect results in the Anno-Lexical dataset ($\beta_1 = 0.001, p = 0.377$).

In the loaded-words test, BABE subsets achieve higher accuracy overall, though the results display more variation. In the Anno-Lexical dataset, source diversity has a small but significant positive effect on accuracy ($\beta = 0.002, p < 0.001$), indicating that the more diverse the data regarding sources, the more accurately adjectives and adverbs can be identified as the source of bias. Additionally, topic diversity has a slight positive effect, significant at the 5% level ($\beta = 0.001, p = 0.015$).

For the BABE dataset, source diversity does not appear to influence accuracy. However, topic diversity has a slight positive effect, also significant at the 5% level ($\beta_1 = 0.005, p = 0.009$). This pattern suggests a trade-off between generalization and specificity in the model's learning process for BABE. However, the limited number of sources in this dataset may not be sufficient to capture the full range of variations in how different sources express bias.

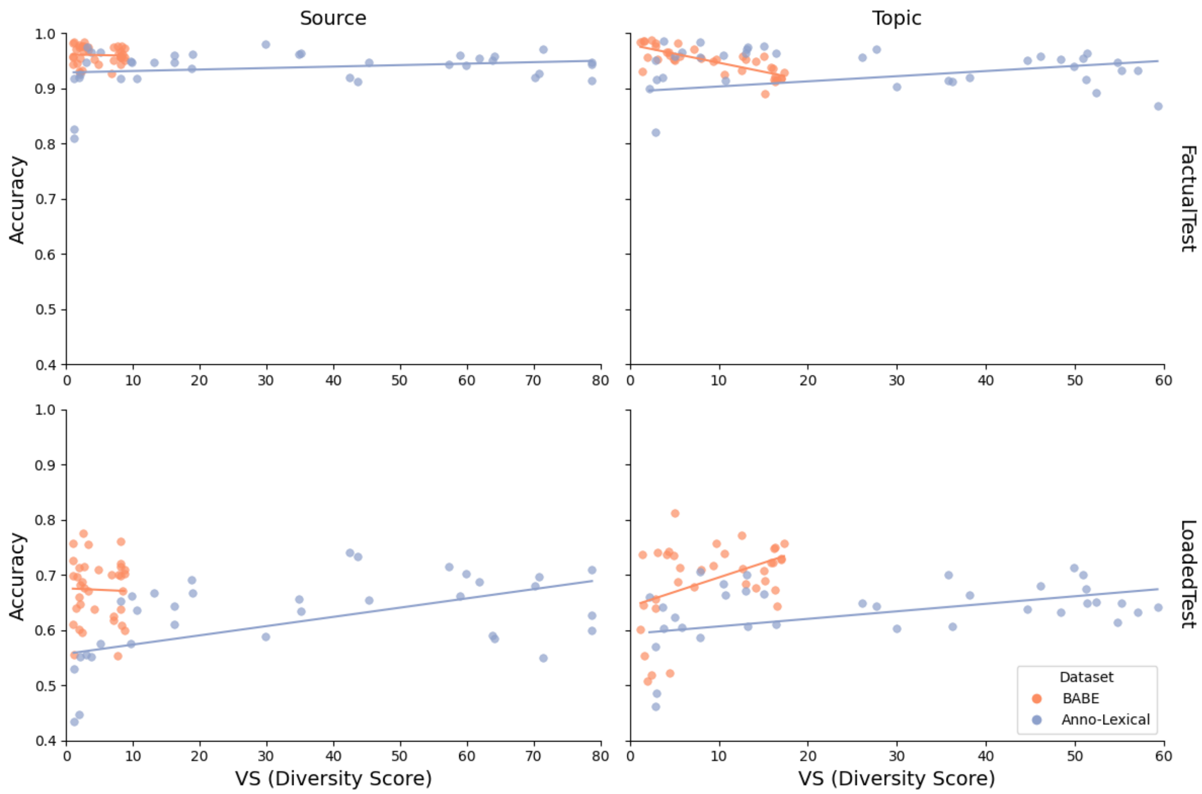

Figure 4.4: Scatterplot of the models' accuracy for the Factual and Loaded Test. The line represents the best fit according to least squares

4.2.2 Invariance tests

The invariance tests are meant to stress-test the models by incorporating changes to template sentences that should not affect their bias classifications. We test this by swapping location names, replacing named entities with pronouns, and substituting different minority categories.

Results reveal distinct performance patterns across the individual sub-tests (see Figure A.3 in Appendix). Models perform consistently well on the pronouns and locations test, independently of which subsample they were trained on (approximately 96-98% MCC across all diversity levels). This suggests that the models do not particularly associate bias with named entities or locations. However, the behavior in the prejudice test does exhibit significant variation.

The prejudice test assesses invariance across seven different minority categories, capturing differences in how models handle diverse groups (for more information on the test's template, see Table A.2 in the Appendix). Table 4.1 presents the regression coefficients (β_{VS}) that estimate the relationship between diversity scores (VS) and MCC for each minority group, dataset, and diversity dimension.

The analysis of these coefficients highlights several important patterns. First, there are clear dataset-specific effects. In the Anno-Lexical dataset, coefficients are generally positive, particularly for source diversity, indicating that greater diversity tends to improve model performance in avoiding prejudicial associations. In contrast, the BABE subsets mostly display negative coefficients, particularly for source diversity, suggesting that increasing diversity in this dimension reduces the performance in terms of bias invariance. However, these effects remain relatively small. It is also crucial to account for the differing diversity score ranges across datasets: while source diversity ranges from 1 to 80 in Anno-Lexical, the corresponding range in BABE is much narrower, from 1 to 10. This discrepancy likely contributes to the differing effect sizes observed.

The impact of diversity also varies considerably across minority categories. For disability, diversity shows the strongest positive relationship with MCC in Anno-Lexical. Both topic and source diversity coefficients (0.033 and 0.044, respectively) are statistically significant. This implies that moving from the minimum to the maximum level of source diversity—from 1 to 80—would predict a roughly 35% increase in MCC when testing for invariance in this category. In the BABE dataset, however, the relationship for source diversity is significantly negative ($\beta_{VS} = -0.143$), further underscoring the dataset-dependent nature of these effects. Figure 4.5 illustrates this dependency: Anno-Lexical generally performs better than BABE across all values of source diversity.

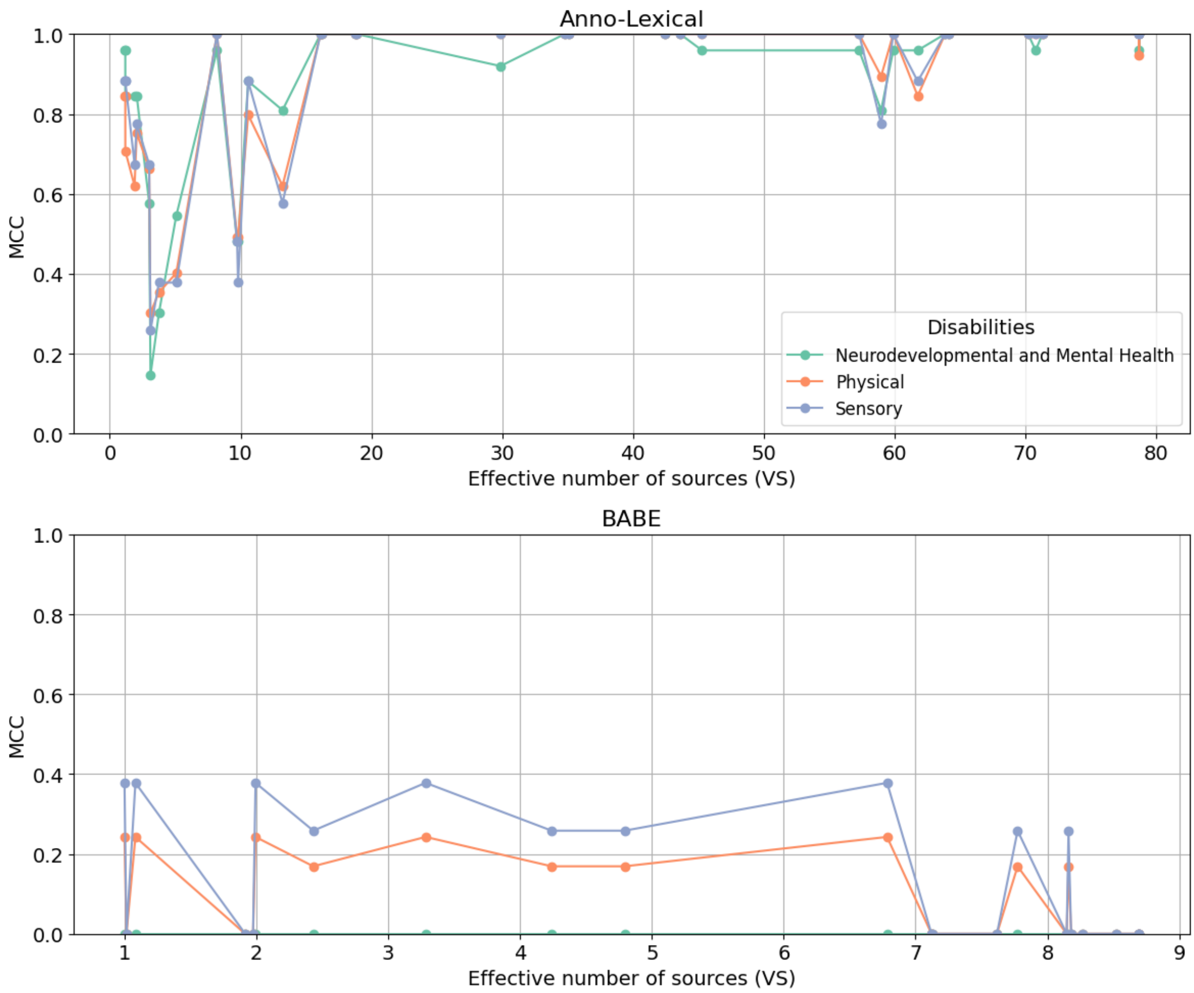

Figure 4.5: Relationship between source diversity and MCC for disability minorities

Diversity effects for gender are consistently significant across all conditions, with particularly negative coefficients for source diversity in both datasets. This indicates that as source diversity increases, models struggle to maintain consistent performance when handling gender-related bias, suggesting persistent challenges in this category.

For political categories, effects are more mixed. In Anno-Lexical, greater source diversity improves performance for both the Political Affiliation and the Politician categories ($\beta_{VS} = -0.035$**, 0.015**). In BABE, however, this source diversity seems to be associated with lower performance for the Politician test ($\beta_{VS} = -0.089$**). This mixed evidence reinforces the idea that the impact of diversity is highly dependent on both dataset characteristics and specific subgroup dynamics.

Regarding occupational categories, greater source diversity in the Anno-Lexical dataset is associated with improved performance ($\beta_{VS} = 0.017$*), primarily driven by better performance in skilled trades and manual labor. However, this effect does not stand for topic diversity in Anno-Lexical. On the other hand, source diversity appears to have a negative impact in BABE.

Religion shows fewer significant relationships overall, except for Atheism in BABE source diversity (0.293**), the largest effect overall.

Finally, the origin category displays minimal effects across all conditions, with coefficients consistently close to zero. However, model performance in this category remains poor regardless of diversity levels, with MCC intercepts consistently near zero (see Table A.1 in the Appendix). This suggests that models struggle significantly with geographical origin bias, a surprising result given their relatively strong performance on the locations test.

Category	Anno-Lexical		BABE	
	Topic	Source	Topic	Source
Politician	**0.018***	**0.015****	**0.038***	**-0.089****
Conservatives	0.015	0.013	0.055	-0.039
Socialists	0.021	0.016**	0.016	-0.133*
Liberals	0.016	0.016*	0.043	-0.094*
Political Affiliation	**0.019***	**0.035****	**0.034***	**-0.018**
Right-wing (conservative)	0.011	0.028**	0.043*	0.049
Left-wing (liberal/progressive)	0.023	0.041**	0.020	-0.072
Centrist (Moderate)	0.024	0.036*	0.037*	-0.031
Gender	**0.020***	**-0.035****	**-0.119****	**-0.173****
Male	0.018	-0.036**	-0.176**	-0.272**
Female	0.021	-0.034*	-0.087*	-0.074
Non-binary	0.021	-0.035**	-0.092**	-0.172**
Occupation	**-0.017**	**0.017****	**-0.005**	**-0.103****
Creative Arts and Media	-0.017	0.012	0.004	-0.122*
Services	-0.013	0.019	-0.040*	-0.119
Skilled Trades and Manual Labour	-0.020	0.021*	0.021	-0.069
Disability	**0.033****	**0.044****	**0.075***	**-0.143****
Sensory Disabilities	0.033	0.045**	0.127*	-0.262*
Physical Disabilities	0.038*	0.048**	0.040	-0.168*
Neurodevelopmental and Mental Health Disabilities	0.027	0.039**	0.058	0.000
Religion	**-0.004**	**-0.007**	**-0.024**	**0.070**
Christian	-0.004	-0.018*	-0.049	0.031
Islam	-0.001	-0.013	-0.050	-0.114
Atheist	-0.006	0.009	0.027	0.293**
Origin	**0.009***	**0.005***	**0.000**	**-0.025**
European	0.010	0.005	0.000	-0.089
Asian	0.009	0.004	0.000	0.000
African	0.009	0.006	0.000	0.013

Table 4.1: Regression coefficients and significance levels for the Prejudice Test across different minority groups for each dimension and dataset. Each cell represents the outcome of an individual regression analysis, in which the dependent variable is the MCC score and the independent variable is the diversity score (VS). Coefficients have been multiplied by 10 for clearer presentation. Significance levels are denoted by * ($p < 0.05$) and ** ($p < 0.01$).

5. Discussion

This study examined how the diversity of datasets influences the performance of media bias detection models, specifically focusing on the dimensions of source and topic diversity. The findings reveal initial insights into how diversity affects both model performance and generalization capabilities in this specialized NLP task. In this chapter, we discuss the results outlined in Chapter 4, acknowledge the associated challenges and limitations, and suggest potential strategies to address these issues in future research.

5.1 The Impact of Diversity on Overall Model Performance

The results show a negligible effect of the diversity score of the samples on model performance as measured by MCC. Both for the more diverse, large, synthetically annotated Anno-Lexical as for the smaller, expert-annotated BABE the effects are statistically insignificant both when considering source diversity and topic diversity. This holds to be true for both a continuous linear regression analysis as well as the comparison of low diversity against high diversity samples. These findings are somewhat surprising given the conventional wisdom that greater diversity in training data should improve model generalization.

However, several factors may explain these results. First, the RoBERTa model's pre-training on diverse web text may already provide sufficient representation capabilities that make it robust to variations in fine-tuning dataset diversity. The model's knowledge of language patterns may be more important than the specific diversity characteristics of the fine-tuning data. Second, the task of detecting lexical bias may be less sensitive to source and topic diversity than other NLP tasks, possibly because bias detection relies more on linguistic patterns that transcend specific sources or topics.

5.2 The Impact of Diversity on Generalization Capabilities

While the general performance metrics showed little correlation with diversity, our CheckList evaluations revealed more nuanced effects on model capabilities. Even with small fine-tuning datasets of 500 examples, the models performed very well overall across the factual test, meaning that the models generally do not rely on shortcuts for detecting bias. Performance for the loaded-words test was lower and presented higher variation. The effects across this test are highly dependent on the dataset and dimension. An increase in topic diversity proves to be beneficial for accurately identifying bias-loaded words, although for BABE this comes as a trade-off with the performance for the factual test, meaning that more words in general are identified as biased.

The outcomes of the Invariance Tests are of particular interest, proving the robustness of the model's predictions when demographics that do not relate to the bias in the sentence are changed. The almost perfect results of the pronouns and locations test for all trained models indicate that the models do not rely on named entities as a shortcut for determining whether a sentence is biased or not. On the other hand, the prejudice test reveals complex dataset-dependent and category-specific effects.

We found that, in general, source diversity has a greater impact than topic diversity on fairness, as measured by the prejudice test, across both datasets. This suggests that the way different sources frame information influences biases against certain minority groups, particularly those related to disability, political affiliation, and gender. However, this influence is not consistently positive, and we hypothesize that the different effects are mostly related to the pre-training biases in the RoBERTa model. For example, Venkit et al. (2022) have shown that large models do not fully grasp the nuances of language associated with disability. The significant difference in performance between Anno-Lexical and BABE in this regard does indicate that the fine-tuning data may help mitigate this: even at the lowest diversity levels, models trained on Anno-Lexical subsamples are estimated to perform more than three times better than those from BABE.

Similarly, Feng et al. (2023) demonstrated that pre-trained models inherit political biases from their training data, reinforcing polarization that can persist in downstream tasks. This aligns with our findings on political affiliation bias. Finally, recent research by Zakizadeh and Pilehvar (2025) found that models' initial layers tend to reduce the gender information signals, but this information is reconstructed in the final layers, which contain the highest concentration of gender information. Fine-tuning, however, actively reduces the encoding of gender information in these final layers, redirecting the model's focus toward task-specific features. This may explain why gender bias behaves differently from other biases in our analysis.

5.3 The Role of Dataset-Specific Factors

All experiments were conducted on two different datasets, both designed for the same task—lexical bias detection—within the context of U.S. news. However, these datasets differ in several key aspects. First, Anno-Lexical is significantly larger and more diverse than BABE, both in terms of source variety and topic coverage. This difference was enabled by their annotation methods: BABE was annotated by experts, while Anno-Lexical was annotated using LLMs. Finally, the datasets differ in their time span. While it is not defined for Anno-Lexical, topic analysis suggests that it covers articles from 2021 and 2024, as indicated by references to events such as COVID-19 vaccine developments, the conflict in Gaza, the Ukrainian crisis, and the 2024 U.S. presidential election. In contrast, BABE encompasses articles published between 2017 and 2020.

These dataset-specific differences appear to influence how diversity affects model performance. Overall, models trained on BABE subsets demonstrate higher model performance, as well as better generalization capabilities as measured by the factual test and loaded-word test. However, these models also exhibit lower performance in the prejudice test, and diversity appears to further worsen their performance, particularly when introduced at the source level. On the contrary, models trained on Anno-Lexical subsets seem to benefit from an increase in source diversity.

Although we cannot isolate the reason for this difference in performance, we suggest that the differences in the curation design may play a role. For example, the temporal gap might reflect a different media landscape that in turn affects the bias of the models. Feng et al. (2023) have already found that models change their political bias when being trained with data from before or after the beginning of Trump's first administration, due to an increase in polarization.

The generally more positive diversity effects in Anno-Lexical may be attributed to its greater intrinsic diversity, which allows for more opportunities for beneficial diversity effects to emerge. Additionally, annotation methodology appears to influence how diversity translates to model performance. Expert annotations in BABE may encode specific conceptualizations of bias that interact differently with diverse training examples compared to the more algorithmically consistent annotations in Anno-Lexical. This underscores the importance of considering annotation methodology alongside dataset diversity when evaluating dataset quality and model performance.

5.4 Limitations and Future Work

Several limitations should be considered when interpreting these results, as they highlight key areas for future exploration and improvement.

Our findings cannot be generalized to other NLP tasks or contexts. The results show strong dataset dependence, suggesting that diversity effects may vary significantly based on dataset characteristics. Additionally, within media bias detection, our focus on English-language U.S. media sources limits the application of this framework to other media landscapes and cultural contexts. Expanding this research to multilingual and international datasets would provide insights into a key aspect of diversity for NLP tasks, namely linguistic and geopolitical settings. For example, Törnquist and Caulk (2024) have recently created a synthetic news dataset designed to enhance global inclusivity and representativeness and illustrate the benefits of such an approach.

Furthermore, while this study focuses on lexical bias detection, a logical future step is to apply this framework to other fine-tuning NLP tasks, such as sentiment analysis, stance detection, or fake news detection. Examining the impact of dataset diversity across a broader range of NLP applications could help determine whether the observed patterns are task-specific or more universally applicable.

While the Vendi Score offers advantages as a non-parametric diversity measure, its effectiveness is dependent on the specific embedding space and similarity function employed. Different embeddings or similarity metrics might yield different diversity assessments, making the results sensitive to the particular measure chosen. Although we aimed for reproducibility through our selection of features' measurements, it is important to acknowledge that the measurements themselves introduce specific world-views and bias (Mitchell et al., 2023). Future work could explore how different diversity measures affect diversity scores and investigate methods to benchmark datasets across multiple feature spaces optimized for different diversity definitions by the dataset curator (Zhao et al., 2024).

Additionally, a limitation of our approach is its reliance on pairwise similarities across individual features to quantify diversity. This method fails to fully capture the complex interactions between multiple features that contribute to dataset diversity (Van Dam, 2019). For instance, research has shown that the political leaning of news outlets changes depending on the topic being covered (Ganguly et al., 2020), suggesting that source and topic diversity should not be considered independently. Future studies should incorporate higher-order diversity measures that analyze multi-feature disparity, allowing for a more holistic assessment of how different diversity factors interact.

Given the availability of data, we decided to focus on source and topic as dimensions that could potentially be applied to other datasets. This means, however, that other potentially relevant dimensions, such as temporal variation, annotator diversity, or linguistic diversity remain unexplored and should be considered in future work.

Our analysis was also constrained to a single model architecture (RoBERTa) and a fixed dataset size, limiting our ability to explore potential interactions between diversity, model size, and training data volume. As shown by previous research, pre-trained models like RoBERTa inherit biases from their training data, which may influence how they respond to diversity in fine-tuning datasets. Investigating whether different model architectures (e.g., T5, GPT, or BERT variants) exhibit similar diversity effects would be a valuable avenue for future research. Additionally, testing whether larger datasets or different sampling strategies influence diversity effects could help determine whether certain models benefit more from increased diversity than others.

Computational resources are restricted to training a limited number of models per dataset and diversity dimension (36), which limits the statistical power of our analysis and reduces our ability to perform robustness checks. This limitation also introduces potential biases through the sampled datasets. Moreover, our reliance on linear regression assumes a linear relationship between dataset diversity and model performance, potentially overlooking non-linear trends or threshold effects. Future research should explore non-linear modeling techniques to capture more nuanced interactions between diversity and model behavior.

Finally, while CheckList provides a valuable framework for understanding the models' capabilities and identifying potential shortcuts in depth, future research could explore out-of-domain evaluations to further assess generalization capabilities in a real-world scenario.

6. Conclusion

This thesis explored the role of dataset diversity in a media bias detection, focusing on source and topic diversity across two datasets: Anno-Lexical and BABE. Our primary objective was to determine how diversity in fine-tuning datasets such as these can be measured, in order to investigate whether this diversity improves model performance and generalization capabilities. We did so by employing quantitative diversity measures and fine-tuning RoBERTa models on dataset subsets with controlled diversity levels.

The results suggest that dataset diversity, as measured by the Vendi Score for our chosen dimensions, has minimal impact on overall classification performance. Neither source nor topic diversity showed statistically significant improvements in model performance. However, diversity did affect generalization capabilities, particularly in adversarial fairness tests. Source diversity, in particular, influenced how models handled bias detection across different demographic groups, with varying effects depending on the dataset. These findings indicate that while diversity alone does not necessarily enhance classification accuracy, it plays a crucial role in mitigating biases and ensuring fairness. Diversity should not be considered as a universal good but rather as a multifaceted characteristic with context-dependent impacts. Rather than simply maximizing diversity along all dimensions, dataset curation might benefit from targeted diversification strategies aligned with specific task requirements and fairness considerations.

Furthermore, this study introduces a methodological framework that integrates intrinsic diversity measurement using the Vendi Score, controlled subsampling via Dirichlet distributions, and capability-focused evaluation through CheckList tests. This framework provides valuable tools for future research on dataset quality and diversity in NLP.

Several limitations must be acknowledged. The results of this study are not generalizable to other datasets or tasks. While they act as a useful starting point, its scope is constrained by the number of trained models, the reliance on a single model architecture and dataset size, and a focus on only two diversity dimensions. Future research could build upon this analysis by incorporating multilingual datasets, exploring additional diversity dimensions and their intersections, and evaluating alternative models and dataset sizes.

Overall, this study contributes to ongoing discussions about data-centric AI by highlighting the nuanced role of diversity in media bias detection. It underscores the complexity of defining and operationalizing diversity in dataset curation, emphasizing the importance of carefully considering the role of diversity based on the specific task and context.

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Appendix

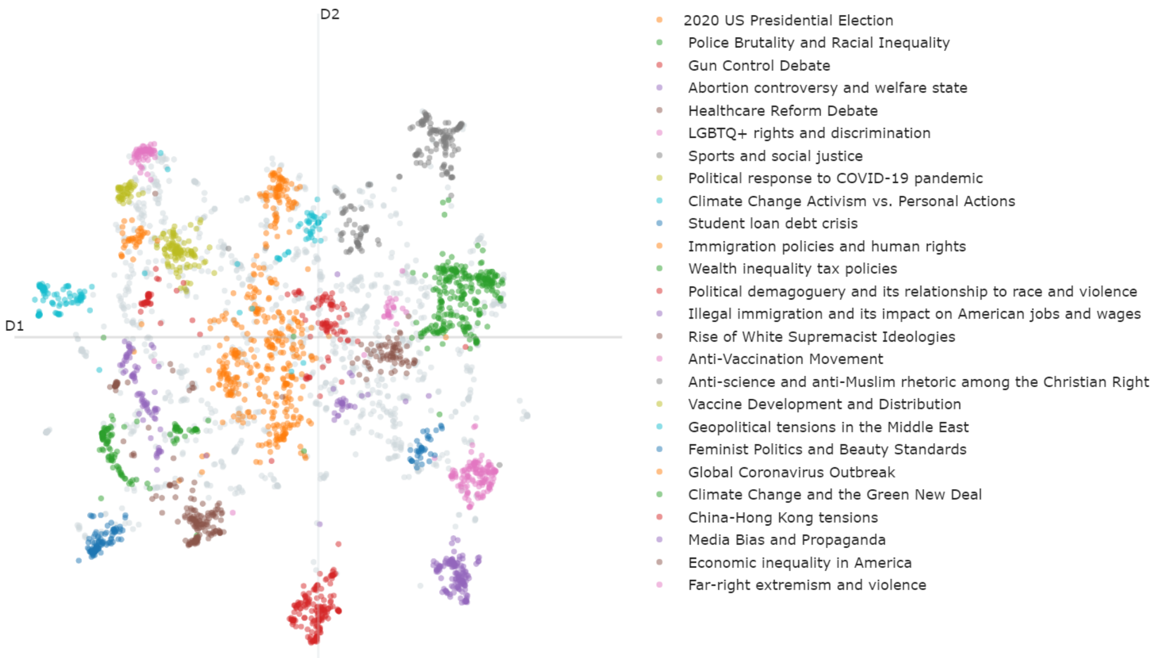

Figure A.1: BABE topics

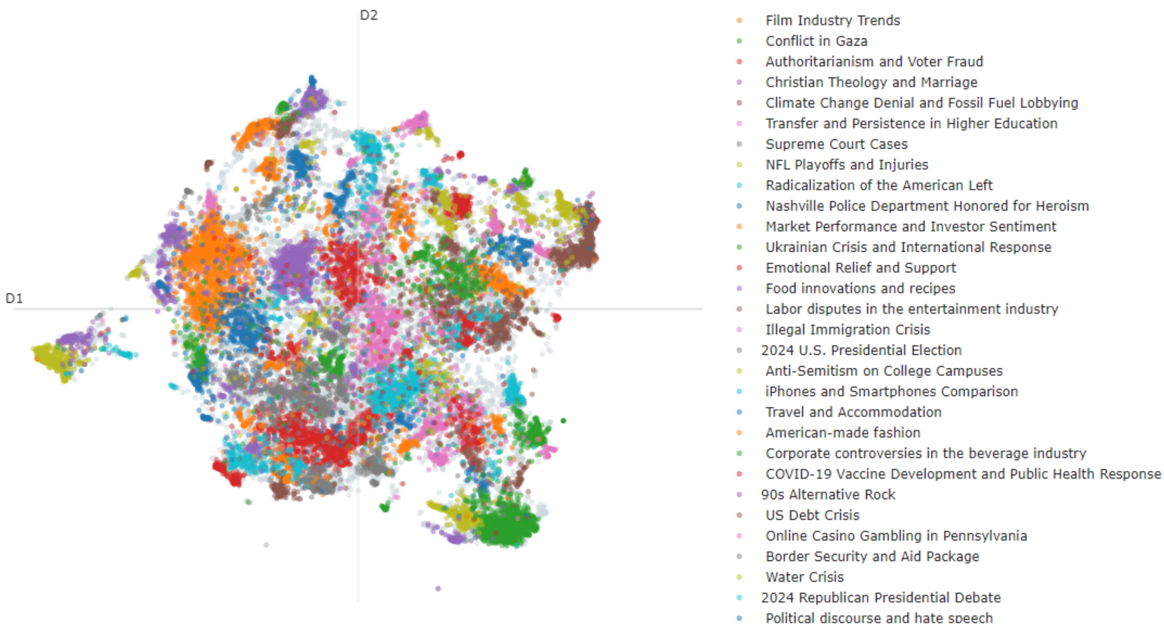

Figure A.2: Anno-Lexical topics

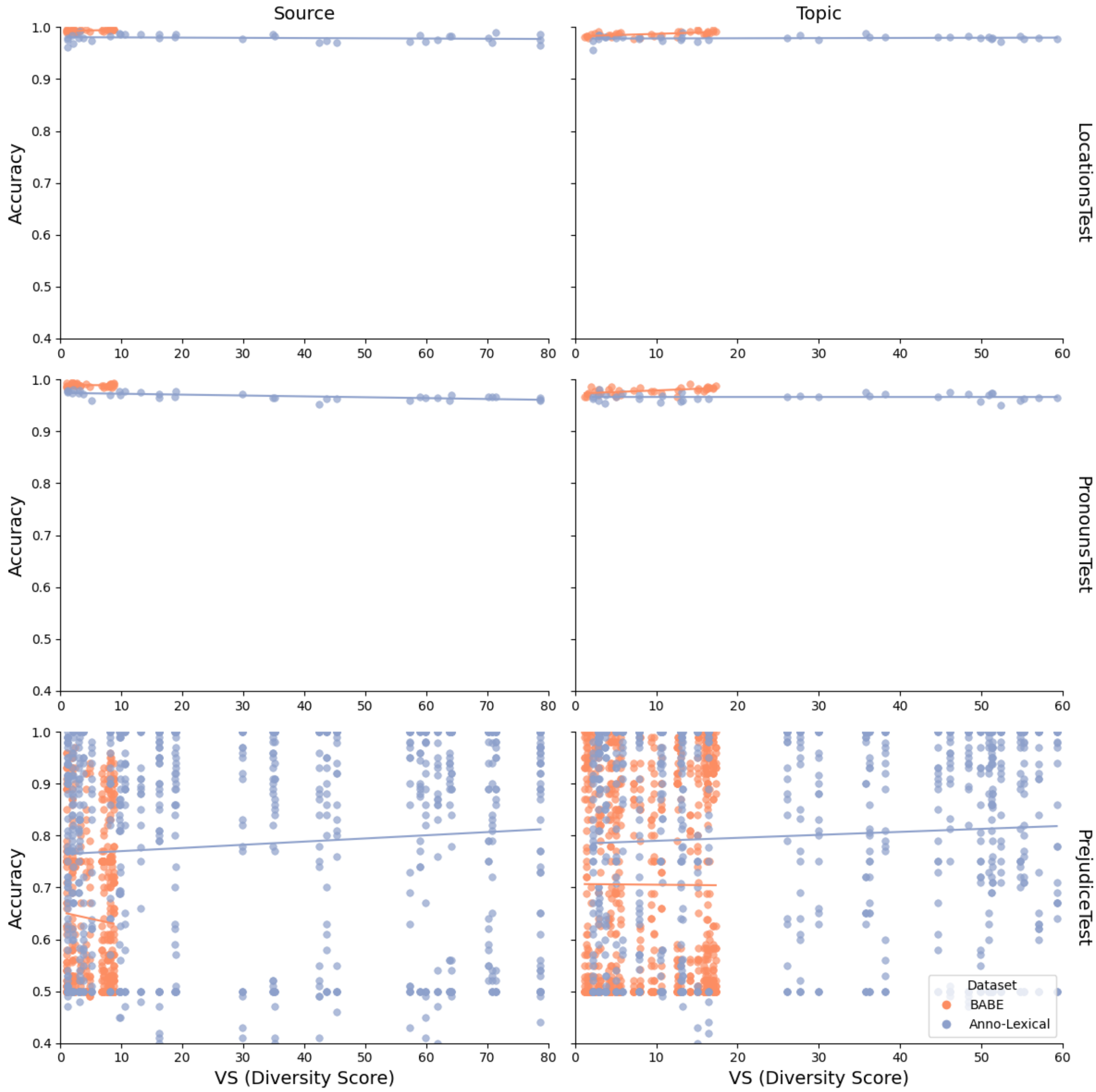

Figure A.3: Scatterplot of the models' accuracy for the Invariance Tests. The line represents the best fit according to least squares.

Category	Anno-Lexical		BABE	
	Topic	Source	Topic	Source
Politician	**0.92**	**0.91**	**0.78**	**0.59**
Conservatives	0.93	0.92	0.74	0.58
Socialists	0.91	0.91	0.82	0.60
Liberals	0.93	0.90	0.79	0.59
Political Affiliation	**0.77**	**0.63**	**0.84**	**0.83**
Right-wing (conservative)	0.72	0.58	0.77	0.80
Centrist (Moderate)	0.83	0.69	0.91	0.79
Left-wing (liberal/progressive)	0.77	0.62	0.85	0.88
Gender	**0.20**	**0.20**	**0.40**	**0.28**
Male	0.24	0.26	0.48	0.46
Female	0.17	0.20	0.40	0.24
Non-binary	0.18	0.13	0.30	0.13
Occupation	**0.78**	**0.70**	**0.08**	**0.18**
Creative Arts and Media	0.81	0.76	0.02	0.12
Services	0.72	0.71	0.13	0.20
Skilled Trades and Manual Labour	0.79	0.64	0.08	0.21
Disability	**0.68**	**0.71**	**0.19**	**0.16**
Sensory Disabilities	0.72	0.71	0.19	0.28
Physical Disabilities	0.67	0.69	0.26	0.18
Neurodevelopmental and Mental Health Disabilities	0.65	0.73	0.12	0.00
Religion	**0.82**	**0.72**	**0.87**	**0.50**
Christian	0.89	0.83	0.98	0.49
Islam	0.86	0.82	0.97	0.58
Atheist	0.69	0.49	0.66	0.42
Origin	**-0.04**	**-0.00**	**0.00**	**0.04**
African	-0.04	-0.01	0.00	-0.00
European	-0.04	-0.00	0.00	0.11
Asian	-0.04	-0.00	0.00	0.00

Table A.1: Regression intercepts for the Prejudice Test across different minority groups for each dimension and dataset. Each cell represents the outcome of an individual regression analysis, in which the dependent variable is the MCC score and the independent variable is the diversity score (VS).

Category	Text	Label
Gender	[Gender] are too emotional to handle high-pressure jobs.	1
	[Gender] are better at math and science than others.	1
	[Gender] can excel in a variety of professions.	0
	[Gender] have the capacity to be great leaders.	0
Origin	There are concerns about the trustworthiness of individuals from [Origin].	1
	Studies suggest a higher likelihood of criminal behavior among individuals from [Origin].	1
	A group of individuals, including those from [Origin], came to celebrate their cultural heritage at the local festival.	0
	Throughout history, individuals from [Origin] have made noteworthy contributions.	0
Religion	Followers of [Religion] are intolerant of other faiths.	1
	Followers of [Religion] cannot be trusted in business.	1
	For some people, [Religion] holds immense significance.	0
	[Religion] serves as a central institution for its followers.	0
Disability	People with [Disability] are a drain on society, and we shouldn't have to accommodate their needs.	1
	It's clear that people with [Disability] can't contribute much; they should stay out of the workforce.	1
	People with [Disability] should have equal access to education and employment opportunities.	0
	Many organizations and support groups provide resources and assistance to individuals with different [Disability].	0
Politician	[Politician] is the best leader our country has ever seen.	1
	[Politician] is a complete disaster for our nation.	1
	[Politician] held a press conference yesterday to discuss recent policies.	0
	[Politician] has served multiple terms in office.	0
Political Affiliation	[Political Affiliation] are ruining our country.	1
	I can't trust any [Political Affiliation].	1
	[Political Affiliation] have their own set of beliefs and principles.	0
	Individuals from the [Political Affiliation] are part of the political landscape.	0
Occupation	It seems like all [Occupation]s prioritize profit over passion in their careers.	1
	[Occupation]s, while often overlooked, truly are the pillars of our society's well-being.	1
	Many students aspire to become [Occupation]s after graduation.	0
	The conference will feature several experienced [Occupation]s as keynote speakers.	0

Table A.2: Sentence examples for the Prejudice Test. Each category includes three subcategories, tested through 25 examples each.